

A brief overview of grading and dg Lie algebras

Grant Talbert

April 21, 2026



Contents

1	Graded structures	1
1.1	Graded R -modules	1
1.2	Operads	5
1.3	Graded associative and commutative R -algebras	7
1.4	Graded rings and their modules	9
1.5	Summary	14
2	Differential graded structures	17
2.1	Differential graded R -modules	17
2.2	Differential graded associative and commutative R -algebras	20
2.3	Differential graded Lie R -algebras	22
2.4	Closure properties	27
2.5	Homological grading conventions	28
2.6	Homology and cohomology	28
3	Differential graded Lie $\mathbb{K}$-algebras	30
3.1	Lie $\mathbb{K}$ -algebra (co)homology	30
3.2	Differential graded Lie $\mathbb{K}$ -algebra (co)homology	32
4	Bundles	37
4.1	Graded bundles	37
4.2	wtf	40

Abstract

In a class I am in we are working through C-G, which requires knowledge of graded structures and dg Lie algebras. My knowledge here is lacking, so I wrote an exposition from my perspective that makes everything clear.

Chapter 1

Graded structures

The definitions of graded structures are often confusing. We show how they arise as natural generalizations of more well-known structures by passing to the functor category $\text{Fun}(\mathbb{Z}, \text{Mod}_R)$ and considering algebras over relevant operads.

1.1 Graded R -modules

Let R be a commutative ring and k a field.

Definition 1.1.1. The category $\mathcal{G}\text{Mod}_R$ of *graded R -modules* is the functor category $\text{Fun}(\mathbb{Z}, \text{Mod}_R)$, where $\mathbb{Z}$ is viewed as a discrete category.

Remark 1.1.2. More explicitly, a graded R -module is a collection $\{M^i\}_{i \in \mathbb{Z}}$ of R -modules, where we write M^i for the functor M evaluated at i . A morphism $f : \{M^i\}_{i \in \mathbb{Z}} \rightarrow \{N^i\}_{i \in \mathbb{Z}}$ of graded R -modules is a collection $f = \{f^i\}_{i \in \mathbb{Z}}$ of morphisms $f^i : M^i \rightarrow N^i$ of R -modules. Note that all universals in Mod_R promote to universals in $\mathcal{G}\text{Mod}_R$, and are computed pointwise. For example, $\{M^i\}_{i \in \mathbb{Z}} \oplus \{N^i\}_{i \in \mathbb{Z}} \cong \{M^i \oplus N^i\}_{i \in \mathbb{Z}}$. Since Mod_R is abelian and $\mathbb{Z}$ is small, we obtain for free that $\mathcal{G}\text{Mod}_R$ is an abelian category.

Terminology 1.1.3. Given a graded R -module $M = \{M^i\}_{i \in \mathbb{Z}}$, we call M^i the i th degree of M . Given a graded R -module M , we write M^i for the i th degree of M . If $x \in M^i$, we say x is a *homogeneous element* of degree i , or simply a *homogeneous element*. We will sometimes write $x \in M$ to denote that x is a homogeneous element of arbitrary or unknown degree. If $x \in M^i$, we say the *degree* of x is i , and write $|x| = i$ to denote the degree of x . If $R = k$ is a field, we call graded k -modules *graded k -vector spaces*. If M is a graded R -module and there exists $i \in \mathbb{Z}$ such that $M^j \cong 0$ for all $j \neq i$, then we say M is *concentrated in degree i* .

Warning 1.1.4. Since Mod_R has tensor product, the functor category $\mathcal{G}\text{Mod}_R$ inherits this tensor product. Thus one might expect that given graded R -modules M, N , we define the tensor product $M \otimes N$ pointwise, meaning $(M \otimes N)^k = M^k \otimes N^k$. Although this indeed yields a symmetric monoidal structure for Mod_R , this is not the correct tensor product to use. We explore how the correct tensor product arises in discussion 1.1.5, construction 1.1.6, and construction 1.1.7.

Discussion 1.1.5. Since graded objects are algebraic structures, it is natural to consider the algebraic structure of $\mathbb{Z}$ when defining graded R -modules. In particular, we are interested in the group structure of $\mathbb{Z}$, since a graded R -module should be seen as a generalization of an R -module, which inherents the structure of an abelian group. Luckily, all groups are in particular monoids, meaning that the discrete category $\mathbb{Z}$ inherits a strict symmetric monoidal structure by taking the

monoidal product to be usual addition. We can now use the *Day convolution*.

Construction 1.1.6. The *Day convolution* takes two symmetric monoidal categories $(\mathcal{C}, \otimes_{\mathcal{C}}), (\mathcal{D}, \otimes_{\mathcal{D}})$ and provides a symmetric monoidal structure on $\text{Fun}(\mathcal{C}, \mathcal{D})$. This requires that the tensor product in $\mathcal{D}$ preserves colimits in both arguments, which follows in Mod_R since it is closed symmetric monoidal.

The Day convolution is canonical in the sense that it yields the unique monoidal structure on $\text{Fun}(\mathcal{C}, \mathcal{D})$ such that Yoneda's embedding $y : \mathcal{C} \rightarrow \text{Fun}(\mathcal{C}^{\text{op}}, \mathcal{D})$ is monoidal. Explicitly, we want $\mathcal{C}(c_1, -) \otimes \mathcal{C}(c_2, -) \cong \mathcal{C}(c_1 \otimes_{\mathcal{C}} c_2, -)$.

To derive the Day convolution, let $F, G : \mathcal{C} \rightarrow \mathcal{D}$. It might be useful to start by considering the functor $F \boxtimes G : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{D}$ given by $(F \boxtimes G)(c_1, c_2) = F(c_1) \otimes_{\mathcal{D}} F(c_2)$. This is functorial since it arises as the composite $\otimes_{\mathcal{D}} \circ (F \times G)$. We want to turn this into a functor $\mathcal{C} \rightarrow \mathcal{D}$ in a natural way. To do this, we can take the left Kan extension

$$\begin{array}{ccc} \mathcal{C} \times \mathcal{C} & \xrightarrow{F \boxtimes G} & \mathcal{D} \\ \searrow \otimes_{\mathcal{C}} & \Downarrow & \nearrow \text{Lan}_{\otimes_{\mathcal{C}}} F \boxtimes G \\ & \mathcal{C} & \end{array}$$

This is clearly a fairly natural way to construct such a structure, since it is the universal way to construct a functor $\mathcal{C} \rightarrow \mathcal{D}$ from F, G which maps $c_1 \otimes_{\mathcal{C}} c_2 \mapsto F(c_1) \otimes_{\mathcal{D}} F(c_2)$. Thus we define $F \otimes D := \text{Lan}_{\otimes_{\mathcal{C}}} F \boxtimes G$.

Now we assume that everything is sufficiently well-behaved such that the Kan extensions exist. We can compute the Kan extension by the coend ([ML98])

$$\begin{aligned} \text{Lan}_{\otimes_{\mathcal{C}}}(F \boxtimes G)(c) &\cong \int^{(c_1, c_2) \in \mathcal{C} \times \mathcal{C}} \mathcal{C}(c_1 \otimes_{\mathcal{C}} c_2, c) \cdot (F \boxtimes G)(c_1, c_2) \\ &\cong \int^{(c_1, c_2) \in \mathcal{C} \times \mathcal{C}} \mathcal{C}(c_1 \otimes_{\mathcal{C}} c_2, c) \cdot (F(c_1) \otimes_{\mathcal{D}} F(c_2)), \end{aligned}$$

where $\cdot$ denotes the copower.

This definition of a tensor product indeed makes $\text{Fun}(\mathcal{C}, \mathcal{D})$ into a symmetric monoidal category. If $1_{\mathcal{C}}$ is the unit for $\otimes_{\mathcal{C}}$, then $\mathcal{C}(1_{\mathcal{C}}, -)$ is the unit for Day's tensor product.

Construction 1.1.7. We now apply discussion 1.1.5 and construction 1.1.6 to construct a tensor product $\otimes$ for $\mathcal{G}\text{Mod}_R$. Since $\mathbb{Z}$ is discrete, the computation of the coend is simplified dramatically by $\mathbb{Z}(i, j)$ being empty when $i \neq j$ and the singleton when $i = j$. Hence for M, N graded R -modules and δ_i^j the Kronecker delta, we have

$$\begin{aligned} (M \otimes N)^k &:= \int^{(i, j) \in \mathbb{Z} \times \mathbb{Z}} \mathbb{Z}(i + j, k) \cdot (M^i \otimes_R M^j) \\ &= \int^{(i, j) \in \mathbb{Z} \times \mathbb{Z}} \delta_{i+j}^k \cdot (M^i \otimes_R M^j). \end{aligned}$$

For convenience of notation, we write $F : (\mathbb{Z} \times \mathbb{Z})^{\text{op}} \times (\mathbb{Z} \times \mathbb{Z}) \rightarrow \mathcal{G}\text{Mod}_R$ for the functor

$$((i, j), (k, \ell)) \mapsto \mathbb{Z}(i + j, -) \cdot (M^k \otimes_R M^{\ell})$$

such that

$$M \otimes N := \int^{\mathbb{Z} \times \mathbb{Z}} F.$$

We can then compute the coend as the coequalizer of the diagram

$$\coprod_{\substack{(f,g) \in \text{Mor}(\mathbb{Z} \times \mathbb{Z}) \\ (f,g):(i,j) \rightarrow (k,\ell)}} F(i,j,k,\ell) \rightrightarrows \coprod_{(i,j) \in \mathbb{Z} \times \mathbb{Z}} F(i,j,i,j)$$

where the arrows act by the action of F on $(f, 1_{(k,\ell)})$ and $(1_{(i,j)}, g)$. Since these morphisms only exist when f, g are the identities, and since F must map an identity to an identity, we see that these actions are trivial here, so both arrows are the identity. Their coequalizer is thus their codomain

$$\coprod_{i,j \in \mathbb{Z}} F(i,j,i,j).$$

Since the coproduct in Mod_R is the direct sum, and by remark 1.1.2, we obtain that this is

$$M \otimes N \cong \bigoplus_{i,j \in \mathbb{Z}} \delta_{i+j}^k (M^i \otimes_R M^j).$$

We now can compute the k th degree of $M \otimes N$:

$$(M \otimes N)^k \cong \bigoplus_{i,j \in \mathbb{Z}} \delta_{i+j}^k (M^i \otimes_R M^j) \cong \bigoplus_{i,j \in \mathbb{Z}}^{i+j=k} M^i \otimes_R M^j.$$

Definition 1.1.8. Let M, N be graded R -modules. Then the *graded tensor product* $M \otimes N$ is defined by

$$(M \otimes N)^k := \bigoplus_{i,j \in \mathbb{Z}}^{i+j=k} M^i \otimes_R M^j.$$

Proposition 1.1.9. The graded tensor product promotes $\mathcal{G}\text{Mod}_R$ to a symmetric monoidal category such that the unit for the tensor is R concentrated in degree 0.

Remark 1.1.10. Let M be an R -module. Then we can view M as a graded R -module by viewing it as the graded R -module concentrated in degree 0 with $M^0 = M$. This yields a canonical embedding $i : \text{Mod}_R \rightarrow \mathcal{G}\text{Mod}_R$ which is easily seen to be fully faithful.

Remark 1.1.11. The category $\mathcal{G}\text{Mod}_R$ is closed symmetric monoidal. To construct the internal hom, we look for a left adjoint $- \otimes M \dashv \underline{\mathcal{G}\text{Mod}}_R(M, -)$. This must satisfy the universal property that for any graded R -modules N, P and any diagram of solid arrows

$$\begin{array}{ccc} N \otimes M & \xrightarrow{f} & P \\ \uparrow & \nearrow \text{dashed} & \\ \underline{\mathcal{G}\text{Mod}}_R(M, N) & & \end{array}$$

there is a unique dashed arrow as indicated, rendering the diagram commutative. To show how this arises, continuity of representable functors and the usual tensor-hom adjunction $- \otimes_R M \dashv \text{Hom}_R(M, -)$ yields

$$\begin{aligned} \mathcal{G}\text{Mod}_R(N \otimes M, P) &\cong \prod_{i,j \in \mathbb{Z}} \text{Hom}_R(N^j \otimes_R M^i, P^{i+j}) \cong \\ &\prod_{i,j \in \mathbb{Z}} \text{Hom}_R(M^i, \text{Hom}_R(N^j, P^{i+j})) \\ &\cong \prod_{i \in \mathbb{Z}} \prod_{j \in \mathbb{Z}} \text{Hom}_R(M^i, \text{Hom}_R(N^j, P^{i+j})) \\ &\cong \prod_{i \in \mathbb{Z}} \text{Hom}_R(M^i, \prod_{j \in \mathbb{Z}} \text{Hom}_R(N^j, P^{i+j})). \end{aligned}$$

An element of this product is a tuple $\{f^i\}_{i \in \mathbb{Z}}$ such that $f^i : M^i \rightarrow \prod_j \text{Hom}_R(N^j, P^{i+j})$. Then simply define

$$\underline{\mathcal{G}\text{Mod}}_R(N, P)_j := \prod_{j \in \mathbb{Z}} \text{Hom}_R(N^j, P^{i+j}).$$

We thus have our adjunction $- \otimes M \dashv \underline{\mathcal{G}\text{Mod}}_R$.

Notation 1.1.12. Throughout section 1.1, we have denoted the tensor product in $\mathcal{G}\text{Mod}_R$ by $\otimes$ and the internal hom by $\underline{\mathcal{G}\text{Mod}}_R$. These notations were very convenient in the short term for making distinctions between operations in $\mathcal{G}\text{Mod}_R$ and Mod_R , but will be inconvenient in the long term. Let M, N be graded R -modules. We will write $M \otimes_R N$ for their graded tensor product and $\underline{\text{Hom}}_R(M, N)$ for their internal hom. We reconcile this notation with the usual use of these notations in Mod_R as follows. Since there is a canonical embedding $\text{Mod}_R \hookrightarrow \mathcal{G}\text{Mod}_R$ by viewing R -modules as concentrated in degree 0, we can consider the graded tensors and internal homs of normal R -modules M, N . By definition 1.1.8,

$$(M \otimes_R N)^k \cong \bigoplus_{i,j \in \mathbb{Z}}^{i+j=k} M^i \otimes_R N^j.$$

Since $M^i \otimes_R N^j$ is nonzero if and only if $i = j = 0$, in which case it is the usual tensor product $M \otimes_R N$, we obtain that the graded tensor product $M \otimes_R N$ is simply the usual tensor product of R -modules, concentrated in degree 0. Hence the inclusion is monoidal. Similarly, since $\text{Hom}_R(M^i, N^j)$ is a singleton whenever i or j is nonzero, the internal hom

$$\underline{\text{Hom}}_R(M, N)_i \cong \prod_{j \in \mathbb{Z}} \text{Hom}_R(M^j, N^{i+j}) \cong \text{Hom}_R(M^{-i}, N^i)$$

is nonzero only in degree 0, where it is simply $\text{Hom}_R(M, N)$. Hence the inclusion also preserves the internal hom.

Warning 1.1.13. One would expect that we would use the symmetry isomorphism $M \otimes_R N \cong N \otimes_R M$ given by $m \otimes n \mapsto n \otimes m$. We *will not use this*. Instead, we will use *Koszul braiding*. This will be important to define various differentiable structures, which is of course our goal, but will have strange effects on the definition of “commutivity” later on. The symmetry isomorphism is defined by $m \otimes n \mapsto (-1)^{|m||n|} n \otimes m$, where $-1 \in R$. We demonstrate one such implication immediately.

Discussion 1.1.14. We consider the evaluation morphism. Given M, N graded R -modules, consider the map $\varepsilon : \underline{\text{Hom}}_R(M, N) \otimes_R M \rightarrow N$ defined as follows. By definition 1.1.8, a homogeneous element of order k is a homogeneous element $f \in \underline{\text{Hom}}_R(M, N)^i$ and $m \in M^j$ with $i + j = k$. By remark 1.1.11, the element f has a component $f^j : M^j \rightarrow N^{i+j}$, so $f^j(m)$ is well-defined and is homogeneous of degree k . We thus define $\varepsilon^k(f \otimes m) = f^{|m|}(m)$.

We explore some implications of this. Let X_1, X_2, Y_1, Y_2 be graded R -modules. The tensor of two morphisms $X_1 \rightarrow X_2$ and $Y_1 \rightarrow Y_2$ should be a degree 0 element of

$$\underline{\text{Hom}}_R(X_1, X_2) \otimes_R \underline{\text{Hom}}_R(Y_1, Y_2).$$

More generally, we can explore the evaluation map for all homogeneous elements of the internal hom. Let $f \in \underline{\text{Hom}}_R(X_1, X_2)$, $g \in \underline{\text{Hom}}_R(Y_1, Y_2)$, $x \in X_1$, and $y \in Y_1$ be homogeneous elements. Define $(f \otimes g)(x \otimes y)$ as the image under the following composite, where σ denotes the symmetry isomorphism:

$$\begin{array}{c} \underline{\text{Hom}}_R(X_1, X_2) \otimes_R \underline{\text{Hom}}_R(Y_1, Y_2) \otimes_R X_1 \otimes_R Y_1 \\ \downarrow 1 \tilde{\otimes} \sigma \otimes 1 \\ \underline{\text{Hom}}_R(X_1, X_2) \otimes_R X_1 \otimes_R \underline{\text{Hom}}_R(Y_1, Y_2) \otimes_R Y_1 \\ \downarrow \varepsilon \otimes \varepsilon \\ X_2 \otimes_R Y_2. \end{array}$$

By warning 1.1.13, we see that

$$(1 \tilde{\otimes} \sigma \tilde{\otimes} 1)(f \otimes g \otimes x \otimes y) = f \otimes (-1)^{|g||x|} x \otimes g \otimes y.$$

Then acting the evaluation map on this yields $(-1)^{|x||g|} f(x) \otimes g(y) \in X_2 \otimes_R Y_2$. Note that this is a degree $|f| + |g| + |x| + |y|$ homogeneous element, the correct degree.

1.2 Operads

Now that we have a notion of a graded R -module, a natural next step is to define a graded R -algebra. The usual notion of an associative R -algebra is an R -module A which is itself a ring, meaning it has a bilinear multiplication map $A \otimes_{\mathbb{Z}} A \rightarrow A$ which is compatible with the R -module structure in the sense of associativity. More precisely, an associative R -algebra is a monoid object in Mod_R . It is useful to at this point view algebras from the perspective of operads.

Definition 1.2.1. Write $\langle n \rangle = \{1, \dots, n\}$. A *colored operad* $\mathcal{O}$ consists of

- A collection $\text{Obj}(\mathcal{O})$ of *objects* or *colors* (we write $x \in \mathcal{O}$ to say $x \in \text{Obj}(\mathcal{O})$);
- For each finite collection $\{x_i\}_{1 \leq i \leq n} \subseteq \mathcal{O}$ and each $y \in \mathcal{O}$, a collection of *n -multimorphisms*

$$\text{Mul}_{\mathcal{O}}(\{x_i\}_{1 \leq i \leq n}, y) = \text{Mul}_{\mathcal{O}}(x_1, \dots, x_n; y).$$

If $f \in \text{Mul}_{\mathcal{O}}(\{x_i\}_{1 \leq i \leq n}, y)$, we write $f : \{x_i\}_{1 \leq i \leq n} \rightarrow y$. If $n = 1$ such that $\{x_i\}_{1 \leq i \leq n} = \{x\}$, then we write $f : x \rightarrow y$ and call f a morphism. We allow $n = 0$, in which case the collection $\{x_i\}_{1 \leq i \leq 0}$ is empty. Morphisms $\emptyset \rightarrow y$ are called 0-multimorphisms.

- For each $x \in \mathcal{O}$, an *identity morphism* $1_x : x \rightarrow x$;

- For each function $f : \langle n \rangle \rightarrow \langle m \rangle$ of finite sets, for every $\{x_i\}_{1 \leq i \leq n} \subseteq \mathcal{O}$, for every $\{y_j\}_{1 \leq j \leq m} \subseteq \mathcal{O}$, and every $z \in \mathcal{O}$, a composition map

$$\left(\prod_{1 \leq j \leq m} \text{Mul}_{\mathcal{O}}(\{x_i\}_{i \in f^{-1}(j)}, y_j) \right) \times \text{Mul}_{\mathcal{O}}(\{y_j\}_{1 \leq j \leq m}, z) \rightarrow \text{Mul}_{\mathcal{O}}(\{x_i\}_{1 \leq i \leq n}, z).$$

This data has to satisfy a convoluted associativity law, and the identity morphisms are left- and right-units for composition in convoluted ways. We refer the reader to [Lur17] definition 2.1.1.1.

Terminology 1.2.2. An *operad* is a colored operad with one color. Given a colored operad $\mathcal{O}$, we may occasionally write $\text{Mul}(\mathcal{O})$ to denote the collection of all multimorphisms in $\mathcal{O}$. We will rarely use this notation beyond definition 1.2.3.

Definition 1.2.3. Let $\mathcal{O}, \mathcal{P}$ be colored operads. A *functor* $F : \mathcal{O} \rightarrow \mathcal{P}$ of colored operads is an object function $F : \text{Obj}(\mathcal{O}) \rightarrow \text{Obj}(\mathcal{P})$ and a multimorphism function $F : \text{Mul}(\mathcal{O}) \rightarrow \text{Mul}(\mathcal{P})$ which respects domain, codomain, composition, and identity in the expected ways. For example, if $f : \{x_i\}_{1 \leq i \leq n} \rightarrow y$ in $\mathcal{O}$, then $F(f) : \{F(x_i)\}_{1 \leq i \leq n} \rightarrow F(y)$ in $\mathcal{P}$.

If $F, G : \mathcal{O} \rightarrow \mathcal{P}$ are functors of colored operads, then a *natural transformation* $\alpha : F \rightarrow G$ is a collection of morphisms $\alpha_x : F(x) \rightarrow G(x)$ for all $x \in \mathcal{O}$ such that, for any $f : \{x_i\}_{1 \leq i \leq n} \rightarrow y$ in $\mathcal{O}$, we have

$$G(f) \circ (\alpha_{x_1}, \dots, \alpha_{x_n}) = \alpha_y \circ F(f).$$

If $n = 0$, then we simply require $G(f) = \alpha_y \circ F(f)$.

Remark 1.2.4. Let $(\mathcal{C}, \otimes)$ be a symmetric monoidal category. Then $\mathcal{C}$ can be regarded as a colored operad by taking, for any $\{x_i\}_{1 \leq i \leq n} \subseteq \mathcal{C}$ and $y \in \mathcal{C}$,

$$\text{Mul}_{\mathcal{C}}(\{x_i\}_{1 \leq i \leq n}, y) := \mathcal{C} \left(\bigotimes_{1 \leq i \leq n} x_i, y \right).$$

Composition is defined as follows. Let $\alpha : \langle n \rangle \rightarrow \langle m \rangle$ be a function of finite sets, and let $\{x_i\}_{1 \leq i \leq n}, \{y_j\}_{1 \leq j \leq m}, z$ be objects in $\mathcal{C}$. Let $f : \{y_j\}_{1 \leq j \leq m} \rightarrow z$ and $\{g_i^j : \bigotimes_i x_i \rightarrow y_j\}_{1 \leq j \leq m}^{i \in \alpha^{-1}(j)}$. Then we define the composite

$$f \circ \{g_{ij}\}_{i \in \alpha^{-1}(j)} := f \circ \bigotimes_{1 \leq j \leq m} g_{ij}.$$

This makes sense since $g_{ij} : \bigotimes_i x_i \rightarrow y_j$, so the map α essentially sits all the g_{ij} s next to each other, maps all of the x_i s into some y_j , and then maps all of them to z under f .

In particular, if $\mathcal{O}$ is a colored operad and $(\mathcal{C}, \otimes)$ is a symmetric monoidal category, we can discuss functors $F : \mathcal{O} \rightarrow \mathcal{C}$.

Intuition 1.2.5. Colored operads are a generalization of the notion of a category in the same sense that multilinear maps generalize linear maps. Indeed, Mod_R forms a colored operad by taking n -multimorphisms to be n -multilinear maps. However, usually it is more useful to consider small colored operads with only one or two colors, and a small number of maps. The idea here is to encode the structure of an algebraic object in a symmetric monoidal category $(\mathcal{C}, \otimes)$ as a functor $\mathcal{O} \rightarrow \mathcal{C}$ for some operad $\mathcal{O}$.

Definition 1.2.6. Let $\mathcal{A}ss$ be the colored operad with one color $*$, one 0-multimorphism $e : \emptyset \rightarrow *$, and one 2-multimorphism $\mu : \{x_1, x_2\} \rightarrow *$ with $x_0 = x_1 = *$, and one n -multimorphism for all $n \geq 2$. Let $\mathcal{C}$ be a symmetric monoidal category. A *associative algebra object* in $\mathcal{C}$ is a functor $A : \mathcal{A}ss \rightarrow \mathcal{C}$.

Discussion 1.2.7. Let $A : \mathcal{A}ss \rightarrow \mathcal{C}$ be an associative algebra object in a symmetric monoidal category $(\mathcal{C}, \otimes, 1)$. Note that the empty tensor product is the unit 1, so the image of the 0-multimorphism under A is $A(e) : 1 \rightarrow A(*)$. The image of the morphism μ is a map $A(\mu) : A(*) \otimes A(*) \rightarrow A(*)$. Suppose that $\mathcal{C} = \text{Mod}_R$, and write $A(*) = A$, $A(e) = e$, and $A(\mu) = \mu$ for simplicity. Then we get a map $e : R \rightarrow A$ and a map $\mu : A \otimes_R A \rightarrow A$. The datum of the map e is a choice of an element $e_A := e(e_R) \in A$ for $e_R \in R$ the identity element. We claim that the composition law implies that $\mu \circ (1_A \otimes e) = 1_A$, for $1_A : A \rightarrow A$ the identity morphism.

Note that $\mathcal{A}ss$ has an identity morphism $1_* : * \rightarrow *$. Consider the function $f : \langle 1 \rangle \rightarrow \langle 2 \rangle$ mapping $1 \mapsto 2$. Definition 1.2.1 yields a composition map

$$(\text{Mul}_{\mathcal{A}ss}(\emptyset, *) \times \text{Mul}_{\mathcal{A}ss}(*, *)) \times \text{Mul}_{\mathcal{A}ss}(*, *) \rightarrow \text{Mul}_{\mathcal{A}ss}(*, *).$$

In particular, this means that composition $\mu \circ (e, 1_*)$ must be 1_* . Now we consider the images under A . First note that $A(1_*) = 1_A$ is the identity morphism, and composition must map to composition. By remark 1.2.4, the composition law says that $1_* \circ (e, 1_*) = 1_* \circ (e \otimes 1_*)$, and thus we obtain $1_* \circ (e \otimes 1_*) = 1_*$ by functoriality. The same argument yields $1_* \circ (1_*, e) = 1_*$. Note that if we take μ to be multiplication and e to be the identity map, this is precisely the definition of an associative R -algebra: R -bilinearity ensures that multiplication is compatible with the R -action, the argument we just did guarantees that the identity element is a unit, and the colored operad axioms guarantee associativity of multiplication; since there is only one n -multimorphism, so all possible composites of μ must yield the same n -ary multiplication operation. More generally, an associative algebra object in $\mathcal{C}$ is precisely a monoid object in $\mathcal{C}$.

Definition 1.2.8. Let $\mathcal{O}$ be a colored operad and $\mathcal{C}$ a symmetric monoidal category. A functor $F : \mathcal{O} \rightarrow \mathcal{C}$ is called an *algebra for $\mathcal{O}$ over $\mathcal{C}$* . A morphism of algebras for $\mathcal{O}$ over $\mathcal{C}$ is a natural transformation. The category of algebras for $\mathcal{O}$ over $\mathcal{C}$ is denoted $\text{Alg}_{\mathcal{O}}(\mathcal{C})$.

Discussion 1.2.9. Let A, B be associative algebra objects in Mod_R . Denote the identity maps by $e_A : R \rightarrow A$ and $e_B : R \rightarrow B$. Denote multiplication by $\mu_A : A \otimes_R A \rightarrow A$ and $\mu_B : B \otimes_R B \rightarrow B$. A morphism $\varphi : A \rightarrow B$ consists of an R -linear map $\varphi : A \rightarrow B$ with some properties. First, we must have $\varphi \circ e_A = e_B$, meaning that φ preserves identity. Then we must have $\varphi \circ \mu_A = \mu_B \circ (\varphi, \varphi)$. By remark 1.2.4, the composite is equal to $\mu_B \circ (\varphi \otimes \varphi)$. This is precisely the definition of preserving multiplication. Thus there is an evident isomorphism $\text{Alg}_{\mathcal{A}ss}(\text{Mod}_R) \cong \text{Alg}_R$.

1.3 Graded associative and commutative R -algebras

Operads will provide a useful perspective throughout our introduction. Our first application is immediate.

Definition 1.3.1. A *graded associative R -algebra* is an algebra for the operad $\mathcal{A}ss$ over $\mathcal{G}\text{Mod}_R$. The category $\mathcal{G}\text{Alg}_R$ of graded associative R -algebras is the category $\text{Alg}_{\mathcal{A}ss}(\mathcal{G}\text{Mod}_R)$.

Discussion 1.3.2. Let A be a graded associative R -algebra. By proposition 1.1.9, the unit for tensor in $\mathcal{G}\text{Mod}_R$ is R concentrated in degree 0, so the unit is a map $e : R \rightarrow A$. The datum of such a map is a homogeneous element $e_A \in A^0$ of degree 0, with the definition $e(r) = re_A$. We will hereafter denote this element as 1 or e . We now investigate the multiplication map. Note by discussion 1.2.7 that for all homogeneous elements $a \in A$, we obtain $a1 = a$.

Discussion 1.2.7 yields a map $\mu : A \otimes_R A \rightarrow A$. In particular, we get maps $\mu^k : (A \otimes_R A)^k \rightarrow A^k$ for all k . By definition 1.1.8, this is equivalently

$$\mu^k : \bigoplus_{i,j \in \mathbb{Z}}^{i+j=k} A^i \otimes_R A^j \rightarrow A^k.$$

For any homogeneous elements $x \in A^i$ and $y \in A^j$ with $i + j = k$, denote the product $\mu^k(x \otimes y)$ by xy . Thus we obtain that $xy \in A^k$, meaning that multiplication can be described as an R -bilinear map sending pairs (x, y) of homogeneous elements of degree i, j , respectively, to a homogeneous element xy of degree $i + j$. There is a convenient way to express this condition. If $U \subseteq A^i$ and $V \subseteq A^j$, then we write

$$UV = \{xy \in A^{i+j} \mid x \in U, y \in V\}.$$

Then $A^i A^j \subseteq A^{i+j}$. Note that this means $A^0 A^0 \subseteq A^0$. In particular, A^0 is an associative R -algebra.

Before we can proceed, we need another notion of operad.

Definition 1.3.3. Let $\mathcal{O}$ be a colored operad. Note that permutations $\sigma : \langle n \rangle \rightarrow \langle n \rangle$ induce composition maps, and hence S_n has a canonical action on n -multimorphisms of $\mathcal{O}$. A *symmetric colored operad* is a colored operad such that composition is equivariant under this action in the relevant sense. A functor of symmetric colored operads is a functor of colored operads that preserves the symmetric group action. Note that all symmetric colored operads and their symmetric functors are also colored operads and their functors in the sense of definition 1.2.1 and definition 1.2.3. A natural transformation of functors of symmetric colored operads is then simply a natural transformation of functors of colored operads.

Example 1.3.4. Let $(\mathcal{C}, \otimes)$ be a symmetric monoidal category with braiding (symmetry isomorphism) σ . Then $\mathcal{C}$ forms a symmetric colored operad by first forming a colored operad as in remark 1.2.4, and then simply noting that by precomposing by tensor powers of σ and the identity morphism, one can obtain an S_n action on multimorphisms of $\mathcal{C}$. For example, the bijection $\langle 2 \rangle \rightarrow \langle 2 \rangle$ given by $1 \mapsto 2$ and $2 \mapsto 1$ induces the map σ^* . Let $f : x \otimes y \rightarrow z$ be a 2-multimorphism. Then $f \circ \sigma : y \otimes x \rightarrow z$ is a map such that the relevant composition maps are equivariant under this action.

Definition 1.3.5. The commutative algebra operad Com is the symmetric algebra with one n -multimorphism for all n and the trivial S_n -action. Let $\mathcal{C}$ be a symmetric monoidal category. A *commutative algebra object* in $\mathcal{C}$ is a functor of symmetric operads $\text{Com} \rightarrow \mathcal{C}$.

Warning 1.3.6. Since Mod_R is a symmetric colored operad with nontrivial symmetric structure, a functor of symmetric operads $\text{Com} \rightarrow \text{Mod}_R$ is *more restrictive* than simply a functor of operads $\text{Com} \rightarrow \text{Mod}_R$, viewing Com as a normal operad. Moreover, a symmetric monoidal category may have multiple symmetry isomorphisms. We will often specify the symmetry isomorphism we are considering.

Definition 1.3.7. A *graded commutative R -algebra* A is a commutative algebra object in $\mathcal{G}\text{Mod}_R$ with respect to the Koszul braiding of warning 1.1.13.

Discussion 1.3.8. We now investigate definition 1.3.7. Let C be a graded commutative R -algebra. Note that Com is precisely the same as Ass when viewed as a normal operad; although we specified the 0- and 2-multimorphisms of Ass , this was purely for pedagogical emphasis. However viewing the operad as symmetric as impacts on the structure of its algebras, as noted in warning 1.3.6. In particular, the multiplication map $\mu : C \otimes_R C \rightarrow C$ must preserve the S_n -action in some sense. We investigate what this means.

Write m for the 2-multimorphism in Com , and $\alpha : \langle 2 \rangle \cong \langle 2 \rangle$ for the permutation $1 \mapsto 2$ and $2 \mapsto 1$. This induces a map $\alpha^* : \text{Mul}_{\text{Com}}((x_1, x_2), y) \rightarrow \text{Mul}_{\text{Com}}((x_2, x_1), y)$. Since S_n acts trivially on Com , we obtain that this action is trivial. This makes sense since there is only one such morphism. In $\mathcal{G}\text{Mod}_R$, by example 1.3.4 we have that the induced map is precomposition by the Koszul braiding σ^* . Since the functor $C : \text{Com} \rightarrow \mathcal{G}\text{Mod}_R$ must preserve this action, we obtain that $\alpha^*(m) \mapsto \mu \circ \sigma$. However, $\alpha^*(m) = m$ since the S_n -action is trivial, and $m \mapsto \mu$. Thus we must have $\mu \circ \sigma = \mu$.

We consider what this means. By the definition of σ given in warning 1.1.13, given homogeneous elements $x, y \in C$, we have that

$$xy = \mu(x \otimes y) = (\mu \circ \sigma)(x \otimes y) = \mu((-1)^{|x||y|} y \otimes x) = (-1)^{|x||y|} yx.$$

This gives us a notion of what we will informally refer to as *graded symmetry*, which means

$$xy = (-1)^{|x||y|} yx.$$

Note that when x, y are homogeneous elements of degree 0, we obtain $xy = yx$. Combined with discussion 1.3.2 says that C^0 is a commutative R -algebra in the usual sense.

Definition 1.3.9. Let $\mathcal{O}$ be a symmetric colored operad and $\mathcal{C}$ a symmetric monoidal category. A *symmetric algebra* for $\mathcal{O}$ over $\mathcal{C}$ is a functor $F : \mathcal{O} \rightarrow \mathcal{C}$, and the category of such algebras is denoted $\mathcal{C}\text{Alg}_{\mathcal{O}}(\mathcal{C})$.

Remark 1.3.10. Given a symmetric colored operad $\mathcal{O}$ and a symmetric monoidal category $\mathcal{C}$, there is an evident functor $\mathcal{C}\text{Alg}_{\mathcal{O}}(\mathcal{C}) \rightarrow \text{Alg}_{\mathcal{O}}(\mathcal{C})$ by viewing $\mathcal{O}$ as a normal colored operad, and its algebras as normal functors of colored operads.

1.4 Graded rings and their modules

We provide one more generalization for everything considered in this section. Since a ring can be defined as a monoid in $(\text{Ab}, \otimes_{\mathbb{Z}})$, it would make sense to define a *graded ring* similarly.

Definition 1.4.1. A *graded ring* is a graded associative $\mathbb{Z}$ -algebra. A *graded commutative ring* is a graded commutative $\mathbb{Z}$ -algebra. The category of graded rings is denoted $\mathcal{G}\text{Ring}$, and the category of graded commutative rings is denoted $\mathcal{GC}\text{Ring}$.

Discussion 1.4.2. As seen in discussion 1.3.2, a graded ring R is simply a collection of abelian groups $\{R^i\}_{i \in \mathbb{Z}}$ such that R^0 has a unit element, and there is a $\mathbb{Z}$ -bilinear multiplication operation mapping $R^i R^j \subseteq R^{i+j}$. As seen in discussion 1.3.8, such a structure is said to be commutative when multiplication satisfies, for any homogeneous elements $x, y \in R$,

$$xy = (-1)^{|x||y|} yx.$$

By discussion 1.3.8, we see that R^0 is a ring in the usual sense.

Definition 1.4.3. Let $\mathcal{O}$ be an operad with a single color $*$. Define the colored operad $\mathcal{O}_{\text{mod}}$ as the 2-colored operad with colors $\mathbf{a}, \mathbf{m}$ and multimorphisms defined as follows. Let $\{x_i\}_{1 \leq i \leq n}, y$ be colors of $\mathcal{O}$.

- If $x_i = y = \mathbf{a}$ for all i , then

$$\text{Mul}_{\mathcal{O}_{\text{mod}}}(\{x_i\}_{1 \leq i \leq n}, y) = \text{Mul}_{\mathcal{O}}(\{*\}^n, *).$$

- If exactly one x_i is $\mathfrak{m}$, the rest are $\mathfrak{a}$, and $y = \mathfrak{m}$, then

$$\text{Mul}_{\mathcal{O}_{\text{mod}}}(\{x_i\}_{1 \leq i \leq n}, y) = \text{Mul}_{\mathcal{O}}(\{*\}^n, *).$$

- All other possible combinations have no n -multimorphisms.

Definition 1.4.4. Let $\mathcal{O}$ be an operad with a single color, $\mathcal{C}$ a symmetric monoidal category, and $F : \mathcal{O}_{\text{mod}} \rightarrow \mathcal{C}$ an algebra for $\mathcal{O}_{\text{mod}}$ over $\mathcal{C}$. Let $A = F(\mathfrak{a})$. Then $F(\mathfrak{m})$ is called an $\mathcal{O}$ -module over A , or simply an A -module object of $\mathcal{C}$. This definition has an evident symmetric version.

Example 1.4.5. Consider a (symmetric) module $F : \text{Com}_{\text{mon}} \rightarrow \text{Ab}$. We first obtain a ring $F(\mathfrak{a})$, which we will denote R , and an abelian group $F(\mathfrak{m})$, which we will denote M . The operadic structure yields maps $R \otimes_{\mathbb{Z}} M \rightarrow M$ and $M \otimes_{\mathbb{Z}} R \rightarrow M$, and the symmetric property requires that they be equal. This is precisely the structure of a (two-sided) R -module. We also see that if we drop the symmetry requirement and instead consider $F : \text{Ass} \rightarrow \text{Ab}$, we get an (R, R) -bimodule. This suggests more definitions.

Definition 1.4.6. Let $\mathcal{O}$ be an operad with a single color. Define the suboperads $\mathcal{LO}_{\text{mod}}, \mathcal{RO}_{\text{mod}} \subseteq \mathcal{O}_{\text{mod}}$ as follows.

- If $x_i = y = \mathfrak{a}$ for all i , then

$$\text{Mul}_{\mathcal{LO}_{\text{mod}}}(\{x_i\}_{1 \leq i \leq n}, y) = \text{Mul}_{\mathcal{O}}(\{*\}^n, *) = \text{Mul}_{\mathcal{RO}_{\text{mod}}}(\{x_i\}_{1 \leq i \leq n}, y).$$

- If $x_n = \mathfrak{m}$, the other x_i 's are $\mathfrak{a}$, and $y = \mathfrak{m}$, then

$$\text{Mul}_{\mathcal{LO}_{\text{mod}}}(\{x_i\}_{1 \leq i \leq n}, y) = \text{Mul}_{\mathcal{O}}(\{*\}^n, *).$$

- If $x_1 = \mathfrak{m}$, the other x_i 's are $\mathfrak{a}$, and $y = \mathfrak{m}$, then

$$\text{Mul}_{\mathcal{RO}_{\text{mod}}}(\{x_i\}_{1 \leq i \leq n}, y) = \text{Mul}_{\mathcal{O}}(\{*\}^n, *).$$

- All other possible combinations have no n -multimorphisms.

Let $F : \mathcal{LO}_{\text{mod}} \rightarrow \mathcal{C}$ be an algebra. Write $A = F(\mathfrak{a})$. Then $F(\mathfrak{m})$ is called a *left $\mathcal{O}$ -module over A* , or simply an A -module object of $\mathcal{C}$. Similarly if $F : \mathcal{RO}_{\text{mod}} \rightarrow \mathcal{C}$, then $F(\mathfrak{m})$ is a *right A -module object* in $\mathcal{C}$.

Example 1.4.7. We can now immediately see that if $F : \mathcal{LAss}_{\text{mod}} \rightarrow \text{Mod}_R$, writing $M = F(\mathfrak{m})$ and $R = F(\mathfrak{a})$, then M is a left R -module: for $n = 2$, we obtain one multiplication map $R \otimes_{\mathbb{Z}} M \rightarrow M$. Since R is a ring, this is precisely a left R -module. Similarly, if $F : \mathcal{RAss}_{\text{mod}} \rightarrow \text{Mod}_R$, with notation as before, then M is a right R -module.

Definition 1.4.8. Let $\mathcal{C}$ be a symmetric monoidal category, $\mathcal{O}$ an operad with one color, and let R be an algebra for $\mathcal{O}$ over $\mathcal{C}$. Define the following. Denote by $\{R\}$ the discrete subcategory of $\text{Alg}_{\mathcal{O}}(\mathcal{C})$ given by R .

- (1) The category $\mathcal{LMod}_R^{\mathcal{O}}(\mathcal{C})$ of *left R -modules* is the pullback $\text{Alg}_{\mathcal{LO}_{\text{mod}}}(\mathcal{C}) \times_{\text{Alg}_{\mathcal{O}}(\mathcal{C})} \{R\}$.
- (2) The category $\mathcal{RMod}_R^{\mathcal{O}}(\mathcal{C})$ of *right R -modules* is the pullback $\text{Alg}_{\mathcal{RO}_{\text{mod}}}(\mathcal{C}) \times_{\text{Alg}_{\mathcal{O}}(\mathcal{C})} \{R\}$.
- (3) The category $\mathcal{BMod}_R^{\mathcal{O}}(\mathcal{C})$ of *R -bimodules* is the pullback $\text{Alg}_{\mathcal{O}_{\text{mod}}}(\mathcal{C}) \times_{\text{Alg}_{\mathcal{O}}(\mathcal{C})} \{R\}$.
- (4) If $\mathcal{O}$ has a symmetric operad structure and A is a functor of symmetric operads, then the category $\text{Mod}_R^{\mathcal{O}}(\mathcal{C})$ of *R -modules* is the pullback $\mathcal{CAlg}_{\mathcal{O}_{\text{mod}}}(\mathcal{C}) \times_{\mathcal{CAlg}_{\mathcal{O}}(\mathcal{C})} \{R\}$.

Discussion 1.4.9. We illustrate definition 1.4.8 in more detail. There are evident inclusion functors $U_{\mathcal{L}} : \mathcal{LO}_{\text{mod}} \rightarrow \mathcal{O}$, $U_{\mathcal{R}} : \mathcal{RO}_{\text{mod}} \rightarrow \mathcal{O}$, and $U : \mathcal{O}_{\text{mod}} \rightarrow \mathcal{O}$, the final of which is a functor of symmetric colored operads whenever $\mathcal{O}$ is symmetric. Precomposition by each of these functors induces restriction morphisms

$$\begin{aligned} \text{Alg}_{\mathcal{LO}_{\text{mod}}}(\mathcal{C}) &\xrightarrow{U_{\mathcal{L}}^*} \text{Alg}_{\mathcal{O}}(\mathcal{C}) \\ \text{Alg}_{\mathcal{RO}_{\text{mod}}}(\mathcal{C}) &\xrightarrow{U_{\mathcal{R}}^*} \text{Alg}_{\mathcal{O}}(\mathcal{C}) \\ \text{Alg}_{\mathcal{BO}_{\text{mod}}}(\mathcal{C}) &\xrightarrow{U^*} \text{Alg}_{\mathcal{O}}(\mathcal{C}) \\ \mathcal{C}\text{Alg}_{\mathcal{O}_{\text{mod}}}(\mathcal{C}) &\xrightarrow{U^*} \mathcal{C}\text{Alg}_{\mathcal{O}}(\mathcal{C}) \end{aligned}$$

which restrict the relevant algebra along the color $\mathfrak{a}$ to obtain an algebra for $\mathcal{O}$ over $\mathcal{C}$. Let $R : \mathcal{O} \rightarrow \mathcal{C}$ be an algebra. Then the pullbacks

$$\begin{array}{ccc} \mathcal{L}\text{Mod}_R^{\mathcal{O}}(\mathcal{C}) & \longrightarrow & \{R\} \\ \downarrow \lrcorner & & \downarrow \\ \text{Alg}_{\mathcal{LO}_{\text{mod}}}(\mathcal{C}) & \xrightarrow{U_{\mathcal{L}}^*} & \text{Alg}_{\mathcal{O}}(\mathcal{C}) \end{array} \quad \begin{array}{ccc} \mathcal{R}\text{Mod}_R^{\mathcal{O}}(\mathcal{C}) & \longrightarrow & \{R\} \\ \downarrow \lrcorner & & \downarrow \\ \text{Alg}_{\mathcal{RO}_{\text{mod}}}(\mathcal{C}) & \xrightarrow{U_{\mathcal{R}}^*} & \text{Alg}_{\mathcal{O}}(\mathcal{C}) \end{array}$$

$$\begin{array}{ccc} \mathcal{B}\text{Mod}_R^{\mathcal{O}}(\mathcal{C}) & \longrightarrow & \{R\} \\ \downarrow \lrcorner & & \downarrow \\ \text{Alg}_{\mathcal{O}_{\text{mod}}}(\mathcal{C}) & \xrightarrow{U^*} & \text{Alg}_{\mathcal{O}}(\mathcal{C}) \end{array} \quad \begin{array}{ccc} \text{Mod}_R^{\mathcal{O}}(\mathcal{C}) & \longrightarrow & \{R\} \\ \downarrow \lrcorner & & \downarrow \\ \mathcal{C}\text{Alg}_{\mathcal{O}_{\text{mod}}}(\mathcal{C}) & \xrightarrow{U^*} & \mathcal{C}\text{Alg}_{\mathcal{O}}(\mathcal{C}) \end{array}$$

can be described explicitly as the largest subcategory of algebras for the relevant module operad such that restricting along the inclusion yields R (or for morphisms of algebras, the component at R must be the identity morphism). For example, $\text{Mod}_R^{\mathcal{O}}(\mathcal{C})$ is the collection of all commutative algebras M for $\mathcal{O}_{\text{mod}}$ such that $M(\mathfrak{a}) = R$, and the collection of all morphisms $f : M \rightarrow N$ of commutative algebras for $\mathcal{O}_{\text{mod}}$ such that $f_{\mathfrak{a}} = 1_R$ is the identity morphism.

Definition 1.4.10. Let R be a graded ring. A graded (left, right, bi) R -module is a (left, right, bi) R -module in the sense of definition 1.4.8, viewing R in the sense of definition 1.4.1. We denote the categories of graded left, right, bi, and regular R -modules, respectively, as

$$\mathcal{GL}\text{Mod}_R, \quad \mathcal{GR}\text{Mod}_R, \quad \mathcal{GB}\text{Mod}_R, \quad \mathcal{G}\text{Mod}_R.$$

Discussion 1.4.11. Let R be a graded ring and F a graded left R -module. By discussion 1.4.9, F is an algebra for $\mathcal{L}\text{Ass}_{\text{mod}}$. To simplify notation, write $F(\mathfrak{m}) = M$ and $F(\mathfrak{a}) = R$. Then as before, we obtain a multiplication map $R \otimes_{\mathbb{Z}} M \rightarrow M$. In view of definition 1.1.8, unpacking the graded tensor product yields that this means $R^i M^j \subseteq M^{i+j}$. More explicitly, if $r \in R^i$ and $m \in M^j$ are homogeneous elements, then $rm \in M^{i+j}$. Note in particular that we have $R^0 M^0 \subseteq M^0$. Since discussion 1.4.2 says that R^0 is a ring, we see that M^0 is an R^0 -module in the usual sense. Right and bimodules all work analogously. By following the logic of discussion 1.2.7 one can also see that the unit $1 \in R^0$ is a left unit for this multiplication in the sense that $1m = m$ for all homogeneous elements $m \in M$.

Now let R be a graded commutative ring and M a graded module. The relationship between the maps $R \otimes_{\mathbb{Z}} M \rightarrow M$ and $M \otimes_{\mathbb{Z}} R \rightarrow M$ is given by the action on 2-multimorphisms induced by

the bijection $\alpha : \langle 2 \rangle \cong \langle 2 \rangle$ given by $1 \mapsto 2$ and $2 \mapsto 1$. We have seen previously that in symmetric monoidal categories, this is given by the symmetry isomorphism. Since we are working in $\mathcal{G}\text{Mod}_{\mathbb{Z}}$, we once again have the Koszul braiding. Following discussion 1.3.8 tells us that if $r \in R$ and $m \in M$ are homogeneous elements, then

$$rm = (-1)^{|r||m|}mr.$$

Note once again that if the degrees of r, m are 0, then $rm = mr$, and M is an R -module in the usual sense. Note also that this gives us a recipe for identifying $\mathcal{GLMod}_R \cong \mathcal{GRMod}_R \cong \mathcal{GMod}_R$ when R is graded commutative: if M is a graded left R -module, then for all homogeneous elements $r \in R, m \in M$, simply define $mr := (-1)^{|r||m|}rm$. This provides M the structure of a graded R -module, and we can do the same for graded right R -modules.

We now investigate morphisms $f : M \rightarrow N$ of graded (left, right, bi) R -modules. By definition 1.4.8, f is a natural transformation of symmetric colored operads, meaning it consists of a pair of maps $f_a : R \rightarrow R$ and $f_m : M \rightarrow N$. We treat the case of left R -modules; the rest follow analogously. Let μ_M, μ_N denote the morphisms $R \otimes_{\mathbb{Z}} M \rightarrow M$ and $R \otimes_{\mathbb{Z}} N \rightarrow N$, respectively. The naturality condition tells us that

$$\mu_M \circ (f_a, f_m) = f_m \circ \mu_N.$$

Let $e_R : \mathbb{Z} \rightarrow R$ be the identity element. Then the naturality condition also tells us that

$$e_R = f_a \circ e_R.$$

Recall from discussion 1.4.9 that f_a must be the identity morphism 1_R , meaning

$$\mu_M \circ (f_a, f_m) = \mu_M \circ (1_R \otimes f_m) = f_m \circ \mu_N.$$

If $r \in R$ and $m \in M$ are homogeneous elements, then this implies that

$$rf_m(m) = \mu_M(r \otimes f_m(m)) = f_m(\mu_N(r \otimes m)) = f_m(rm).$$

Note that in the case that R, M are concentrated in degree 0, this boils down to R -linearity in the usual sense.

Terminology 1.4.12. Let R be an algebra for an operad with one color in a symmetric monoidal category. Whenever we say M is a (left, right, bi) R -module, we always abuse notation and write M for $M(\mathbf{m})$. Similarly, if $f : M \rightarrow N$ is a morphism of (left, right, bi) R -modules, then we will always abuse notation and write f for $f_{\mathbf{m}}$.

Remark 1.4.13. If R is a ring, we may view it as a graded ring by concentrating it in degree 0. If M is a graded (left, right, bi) R -module, then each M^i is then a (left, right, bi) R -module in the usual sense. For example, if M is a graded left R -module, then the grading says $R^0 M^i \subseteq M^i$ for all i , and thus $R^0 = R$ has a left $\mathbb{Z}$ -bilinear action on M^i . In particular, if R is a graded ring, then there is no ambiguity with the notation Mod_R .

Construction 1.4.14. Let R be a graded commutative ring. The category $\mathcal{GMod}_R$ also has a graded tensor product and an internal hom, which allows all other definitions formulated in this chapter to be extended to graded modules over graded commutative rings. We begin by constructing the graded tensor product.

We start by working in the symmetric monoidal category $(\mathcal{GMod}_{\mathbb{Z}}, \otimes_{\mathbb{Z}}, \mathbb{Z})$ with $\mathbb{Z}$ concentrated in degree 0, and $\otimes_{\mathbb{Z}}$ the Day tensor product as defined in definition 1.1.8. We as always use

Koszul's braiding isomorphism for symmetry. Let M, N be graded R -modules, and write the left and right actions as $\mu_M : M \otimes_{\mathbb{Z}} R \rightarrow M$ and $\mu_N : R \otimes_{\mathbb{Z}} N \rightarrow N$. One characterization of the tensor product in linear algebra is as the coequalizer

$$M \otimes_R N \cong \text{coeq} \left(M \otimes_{\mathbb{Z}} R \otimes_{\mathbb{Z}} N \xrightarrow[1_M \otimes \mu_N]{\mu_M \otimes 1_N} M \otimes_{\mathbb{Z}} N \right),$$

which makes sense since the tensor product should be universal with respect to setting $mr \otimes n = m \otimes rn$ for $m \in M$, $r \in R$, and $n \in N$, and that is precisely what this coequalizer is. We thus choose to use this definition to generalize the tensor product to graded rings.

Recall that since $\mathbb{Z}$ is a normal ring, $\mathcal{G}\text{Mod}_{\mathbb{Z}}$ is a functor category, so appealing to remark 1.1.2 yields that the coequalizer may be computed pointwise. Hence we obtain

$$(M \otimes_R N)^k = \text{coeq} \left(\bigoplus_{i,j,d \in \mathbb{Z}}^{i+j+d=k} M^i \otimes_{\mathbb{Z}} R^d \otimes_{\mathbb{Z}} N^j \xrightarrow[1_M \otimes \mu_N]{\mu_M \otimes 1_N} \bigoplus_{p,q \in \mathbb{Z}}^{p+q=k} M^p \otimes_{\mathbb{Z}} N^q \right).$$

We may now appeal to the coequalizer in the category of normal $\mathbb{Z}$ -modules, which is constructed as the quotient of the codomain by the image of the difference $\mu_M \otimes 1_N - 1_M \otimes \mu_N$. The image can be described as all elements of the form $mr \otimes n - m \otimes rn$, where $m \in M$, $r \in R$, and $n \in N$ are homogeneous elements whose degrees sum to k . Thus the image is the direct sum

$$\bigoplus_{i,j,d \in \mathbb{Z}}^{i+j+d=k} M^i R^d \otimes_{\mathbb{Z}} N^j - M^i \otimes_{\mathbb{Z}} R^d N^j.$$

Thus we obtain

$$(M \otimes_R N)^k := \frac{\bigoplus_{p,q \in \mathbb{Z}} M^p \otimes_{\mathbb{Z}} N^q}{\bigoplus_{i,j,d \in \mathbb{Z}}^{i+j+d=k} M^i R^d \otimes_{\mathbb{Z}} N^j - M^i \otimes_{\mathbb{Z}} R^d N^j}.$$

In view of discussion 1.4.11, the coequalizer now forces the relation

$$rm \otimes n = (-1)^{|r||m|} mr \otimes n = (-1)^{|r||m|} m \otimes rn.$$

Remark 1.4.15. Note that in construction 1.4.14, if R is concentrated in degree 0, then we are quotienting by precisely R -bilinearity, meaning that we perfectly recover the usual graded tensor product of R -modules.

Remark 1.4.16. As with normal linear algebra, we can construct a tensor product of left and right R -modules when R is a not necessarily commutative graded ring. However, such a structure is not naturally an R -module, and is only a $\mathbb{Z}$ -module. Hence we generally only work with graded commutative rings. This is more than sufficient generality, since most of our applications in Lie theory will be over one of the fields $\mathbb{R}$ or $\mathbb{C}$, concentrated in degree 0.

Discussion 1.4.17. We can also construct the internal hom $\underline{\text{Hom}}_R(M, N)$ for a graded commutative ring R and graded R -modules M, N . We will not detail the proof, but we claim that the internal hom can be written as the equalizer

$$\underline{\text{Hom}}_R(M, N) \cong \text{eq} \left(\underline{\text{Hom}}_{\mathbb{Z}}(M, N) \xrightarrow[\beta]{\alpha} \underline{\text{Hom}}_{\mathbb{Z}}(R \otimes_{\mathbb{Z}} M, N) \right),$$

where α, β are defined as

$$\alpha(f)(r \otimes m) = f(rm), \quad \beta(f)(r \otimes m) = rf(m).$$

One can see that this is the case by using continuity and contravariance of Hom in the first argument, and applying the tensor-hom adjunction in $\mathbb{Z}$ to a similar argument to that of remark 1.1.11. The equalizer is the largest subset of the domain which is equalized by these two maps. A homogeneous element f of degree i is thus a tuple $\{f^j : M^j \rightarrow N^{i+j}\}_{j \in \mathbb{Z}}$ of $\mathbb{Z}$ -linear maps which respects the R -action in the sense that

$$f^j(rm) = (-1)^{|r||f|} f^{j-|r|}(m).$$

It is important to remark that a degree 0 homogeneous element f is a graded $\mathbb{Z}$ -module morphism $f : M \rightarrow N$ which satisfies $f(rm) = rf(m)$ for all homogeneous elements $r \in R$ and $m \in M$. Note that this is precisely a morphism of graded R -modules, hence obtain

$$\underline{\text{Hom}}_R(M, N)^0 = \text{Hom}_R(M, N).$$

We could also show this from the tensor-hom adjunction.

Remark 1.4.18. Similarly to warning 1.1.13, when working in the category $\mathcal{G}\text{Mod}_R$ with R a graded commutative ring, we use the symmetry isomorphism $M \otimes_R N \cong N \otimes_R M$ given by $m \otimes n \mapsto (-1)^{|m||n|} n \otimes m$. We refer to this as Koszul braiding.

Remark 1.4.19. Let R be a graded ring. Since the categories of (left, right, bi) R -modules are all defined as specific functor categories, all (co)limits that exist in them are computed pointwise. Since the category is constant at the fiber of R , we see that computing (co)limits pointwise means computing (co)limits on the actual module, which is an object in $\text{Mod}_{\mathbb{Z}}$, which is also a functor category. Remark 1.1.2 now gives us that Mod_R is an abelian category for free.

We have now developed sufficient machinery to re-characterize all constructions in this chapter over graded R -modules for graded rings. For example, a graded associative R -algebra A is an algebra for Ass in $\mathcal{G}\text{Mod}_R$, which can be described as a graded R -module with a multiplication $A \otimes_R A \rightarrow A$ which is compatible with the R -action in the sense that, for $r \in R$ and $a, b \in A$ homogeneous elements,

$$rab = (-1)^{|a||r|} arb.$$

1.5 Summary

We provide a summary of important definitions, constructions, and results from this chapter. Let R be a commutative ring.

1. A graded R -module is a collection $M = \{M^i\}_{i \in \mathbb{Z}}$ of R -modules M^i . A morphism $f : M \rightarrow N$ of graded R -modules is a collection $\{f^i : M^i \rightarrow N^i\}_{i \in \mathbb{Z}}$ of R -linear maps. The category of graded R -modules is denoted $\mathcal{G}\text{Mod}_R$.
2. The graded tensor product $M \otimes_R N$ of graded R -modules M, N is defined as the graded R -module

$$(M \otimes_R N)^k := \bigoplus_{\substack{i+j=k \\ n \in \mathbb{Z}}} M^i \otimes_R N^j,$$

where $M^i \otimes_R N^j$ is the usual tensor product of R -modules. This yields a symmetric monoidal structure for $\mathcal{G}\text{Mod}_R$. The symmetry isomorphism $M \otimes_R N \cong N \otimes_R M$ is defined by $m \otimes n \mapsto (-1)^{|m||n|} n \otimes m$, where $m \in M$ and $n \in N$ are homogeneous elements.

3. The category $\mathcal{G}\text{Mod}_R$ is closed monoidal, with internal hom

$$\underline{\text{Hom}}_R(M, N)^i := \prod_{j \in \mathbb{Z}} \text{Hom}_R(M^j, N^{i+j}).$$

Note that $\underline{\text{Hom}}_R(M, N) \cong \text{Hom}_R(M, N)$.

4. There is a canonical embedding $\text{Mod}_R \hookrightarrow \mathcal{G}\text{Mod}_R$ by viewing R -modules as concentrated in degree 0. This embedding is strict monoidal. This embedding also preserves the internal hom.
5. A graded R -algebra is a graded R -module A together with a map $\pi : A \otimes_R A \rightarrow A$ called multiplication, and denoted $m(x, y) = xy$ for $x, y \in A$ homogeneous elements. The product xy is a homogeneous element of degree $|x| + |y|$. This implies that $A^i A^j \subseteq A^{i+j}$ for all $i, j \in \mathbb{Z}$. A morphism $f : A \rightarrow B$ of graded R -algebras is a morphism of graded R -modules which commutes with multiplication in the usual way. The category of graded R -algebras is denoted $\mathcal{G}\text{Alg}_R$. The zeroth degree A^0 is an associative R -algebra.
6. A commutative graded R -algebra A is a graded R -algebra A which satisfies the property that, for any homogeneous elements $x, y \in A$,

$$xy = (-1)^{|x||y|}yx.$$

A morphism of commutative graded R -algebras is a morphism of graded R -algebras. The category of commutative graded R -algebras is denoted $\mathcal{C}\mathcal{G}\text{Alg}_R$. The zeroth degree A^0 is a commutative R -algebra.

7. There are canonical embeddings $\text{Alg}_R \hookrightarrow \mathcal{G}\text{Alg}_R$ and $\mathcal{C}\text{Alg}_R \hookrightarrow \mathcal{C}\mathcal{G}\text{Alg}_R$ by viewing (associative, commutative) R -algebras as concentrated in degree 0.
8. A graded ring R is a graded $\mathbb{Z}$ -algebra, meaning it is a collection $\{R^i\}_{i \in \mathbb{Z}}$ of abelian groups with a multiplication operation $R \otimes_{\mathbb{Z}} R \rightarrow R$ satisfying $R^i R^j \subseteq R^{i+j}$ for all $i, j \in \mathbb{Z}$. A morphism of graded rings is a morphism of graded $\mathbb{Z}$ -algebras. The category of graded rings is denoted $\mathcal{G}\text{Ring}$. The zeroth degree R^0 is a ring.
9. A commutative graded ring R is a commutative graded R -algebra. The category of commutative graded rings is denoted $\mathcal{C}\mathcal{G}\text{Ring}$. The zeroth degree R^0 is a commutative ring.

We can repeat all of this over graded rings. When a graded ring is concentrated in degree 0, we recover everything here. Let R be a graded commutative ring.

1. A graded R -module M is a graded $\mathbb{Z}$ -module together with a multiplication operation $R \otimes_{\mathbb{Z}} M \rightarrow M$. This means for any homogeneous elements $r \in R$ and $m \in M$, we have $rm \in M^{|r|+|m|}$. More precisely, $R^i M^j \subseteq M^{i+j}$. We also define $rm = (-1)^{|r||m|}mr$.
2. A morphism $f : M \rightarrow N$ of graded R -modules is a morphism of graded $\mathbb{Z}$ -modules which preserves the R -action in the sense that, for any homogeneous elements $r \in R^i$ and $m \in M^j$, we have $f^{i+j}(rm) = r f^j(m)$. The category of graded R -modules is denoted $\mathcal{G}\text{Mod}_R$.
3. The category $\mathcal{G}\text{Mod}_R$ has a symmetric monoidal structure with graded tensor product given by, for M, N graded R -modules,

$$(M \otimes_R N)^k := \frac{\bigoplus_{p,q \in \mathbb{Z}} M^p \otimes_{\mathbb{Z}} N^q}{\bigoplus_{i,j,d \in \mathbb{Z}} M^i R^d \otimes_{\mathbb{Z}} N^j - M^i \otimes_{\mathbb{Z}} R^d N^j}.$$

The symmetry isomorphism in $\mathcal{G}\text{Mod}_R$ is given by $m \otimes n \mapsto (-1)^{|m||n|} n \otimes m$.

4. The category $\mathcal{G}\text{Mod}_R$ is closed monoidal, with internal hom given by

$$\underline{\text{Hom}}_R(M, N)^i := \left\{ f \in \underline{\text{Hom}}_{\mathbb{Z}}(M, N)^i \mid f^j(rm) = (-1)^{|r||f|} f^{j-|r|}(m) \right\}.$$

Here we note that in the above expression, $|f| = i$. Note that $\underline{\text{Hom}}_R(M, N)^0 = \text{Hom}_R(M, N)$.

Chapter 2

Differential graded structures

Generally when we have some kind of graded structure, it is useful to include a differential. The constructions of chapter 1 can immediately be re-defined in a differential graded sense, but some of the rules seem arbitrary. They are clarified by recasting in a categorical light. For maximal generality, we always let R be a graded ring, but most interesting examples occur when R is a regular ring. The notions coincide by viewing a normal ring as a graded ring concentrated in degree 0.

2.1 Differential graded R -modules

Definition 2.1.1. For all $n \in \mathbb{Z}$, define the functor $\mathcal{G}\text{Mod}_R \rightarrow \mathcal{G}\text{Mod}_R$ given by $M \mapsto M[n]$, where $M[n]^i = M^{i+n}$. The action on morphisms is as follows. If $f : M \rightarrow N$ is a morphism of graded R -modules, then $f[n] : M[n] \rightarrow N[n]$ is the morphism of graded R -modules $f[n]^i = f^{i+n}$. We will often omit the $[n]$ on morphisms; for example given $f : M \rightarrow N$ and $g : N[-1] \rightarrow P$, we will write $g \circ f$ to denote the composite $g[1] \circ f$ or $g \circ f[-1]$, depending on the context.

Terminology 2.1.2. Let M be a graded R -module and $n \in \mathbb{Z}$. A map $f : M \rightarrow M[n]$ is said to have *cohomological degree* n .

Definition 2.1.3. A *differential graded R -module* (M, d) is a graded R -module M together with a map $d : M[-1] \rightarrow M$ of cohomological degree 1 known as the *differential*, which has the property that $d \circ d = 0$, where $d \circ d$ denotes the composite

$$M[-1] \xrightarrow{d} M \xrightarrow{d[1]} M[1].$$

A differential graded R -module is also called a cochain complex (with values in R). A *morphism* $f : (M, d_M) \rightarrow (N, d_N)$ of differential graded R -modules is a morphism $f : M \rightarrow N$ of graded R -modules such that $d_N \circ f[-1] = f \circ d_M$. The category of differential graded R -modules is denoted $\text{Ch}(R)$.

Discussion 2.1.4. Let (M, d) be a differential graded R -module. Since $M[1]^i = M^{i+1}$, this means we have maps $d^i : M^{i-1} \rightarrow M^i$. The property that $d[1] \circ d = 0$ implies that $d^{i+1} \circ d^i = 0$ for all i . We usually abbreviate this as $d^2 = 0$, unless there is possibility of confusing the double composite d^2 with the second degree map d^2 , in which case we specify.

Now let (M, d_M) and (N, d_N) be differential graded R -modules and $f : M \rightarrow N$ a morphism of differential graded R -modules. Write $f^i : M^i \rightarrow N^i$ for the i th degree of the map. The condition that f be a morphism can be spelled out more explicitly as $f^i \circ d_M^i = d_N^i \circ f^{i-1}$.

Example 2.1.5. Any graded R -module can be turned into a differential graded R -module by taking

all of the differentials to be 0. In particular, we consider R as a differential graded R -module (R, d_R) by taking $d_R = 0$. By convention, we always take R to be a differential graded R -module in this way.

Other canonical examples are as follows. Given a topological space X and a ring R , the singular cochain complex $C^\bullet(X; R)$ with values in R is a differential graded R -algebra. Given a smooth manifold M , the de Rham complex $\Omega^\bullet(M)$ is a differential graded $\mathbb{R}$ -algebra.

Discussion 2.1.6. The functors $[n]$ defined in definition 2.1.1 extend to functors $\mathcal{Ch}(R) \rightarrow \mathcal{Ch}(R)$. It is convention to define, if (M, d_M) is a differential graded R -module, the differential $d_{M[n]} := (-1)^n d_M[n]$.

Remark 2.1.7. Recall from discussion 1.4.17 that $\mathcal{GMod}_R$ is closed symmetric monoidal, and if R is concentrated in degree 0, then remark 1.1.11 yields the internal hom

$$\underline{\mathrm{Hom}}_R(M, N)^i = \prod_{j \in \mathbb{Z}} \mathrm{Hom}_R(M^j, N^{i+j}).$$

Here we implicitly write Hom_R for the hom-set in the category Mod_R whenever the arguments are normal modules. Note that a homogeneous element of degree i is a tuple of maps $\{f^j : M^j \rightarrow N^{i+j}\}_{j \in \mathbb{Z}}$, which is precisely a morphism of graded R -modules $f : M \rightarrow N[i]$. Hence we can write

$$\underline{\mathrm{Hom}}_R(M, N)^i = \mathrm{Hom}_R(M, N[i]).$$

Thus a homogeneous element of degree i of the internal hom for $\mathcal{GMod}_R$ is a graded R -module map of cohomological degree i .

Our goal for now is to promote the tensor product to $\mathcal{Ch}(R)$. The definition of the differential is done most naturally in the context of the internal hom for $\mathcal{Ch}(R)$.

Construction 2.1.8. We want to use remark 2.1.7 to construct an internal hom for $\mathcal{Ch}(R)$ in a similar sense to remark 1.1.11. For now, we will write the internal hom for $\mathcal{Ch}(R)$ as $\underline{\mathcal{Ch}}_R(-, -)$, and the normal hom as $\mathcal{Ch}_R(-, -)$.

Let $(M, d_M), (N, d_N)$ be differential graded R -modules. Since we want the tensor product to promote to $\mathcal{Ch}(R)$, it makes sense to define $\underline{\mathcal{Ch}}_R(M, N) = \underline{\mathrm{Hom}}_R(M, N)$, and to simply define the differentials. Initially this seems strange since $\underline{\mathcal{Ch}}_R(M, N)$ now contains information that does not respect the differential (for example, one might expect $\mathcal{Ch}_R(M, N)$ to be the degree 0 homogeneous elements). However, the internal hom just needs to respect the canonical identification

$$\mathcal{Ch}_R(M, N) \cong \mathcal{Ch}_R(R \otimes_R M, N) \cong \mathcal{Ch}_R(R, \underline{\mathcal{Ch}}_R(M, N)).$$

Thus as long as we define the differentials in the correct way, we can still extract the proper data from this definition.

We explore this identification further. Recall by discussion 1.4.17 that degree 0 homogeneous elements of the internal hom are $\mathrm{Hom}_R(M, N)$. Let $f : R \rightarrow \underline{\mathcal{Ch}}_R(M, N)$, and consider the degree 0 morphism $f^0 : R^0 \rightarrow \mathrm{Hom}_R(M, N)$. Since R^0 is a ring, f^0 is uniquely determined by the image of its identity element $1 \in R^0$. Since f is a morphism of differential graded R -modules, we obtain $f^1 \circ d_R^0 = d^0 \circ f^0$, where d^0 is the zeroth degree of the differential on the internal hom. Since $d_R^0 = 0$, this equation reads $d^0 \circ f^0 = 0$. In particular, $(d^0 \circ f^0)(1) = 0$. Let $\varphi = f^0(1) : M \rightarrow N$. Recall that we want the possible choices of φ to correspond exactly to differential graded R -module homomorphisms, so choose φ to be differential graded, meaning $d_N \circ \varphi = \varphi \circ d_M$. In particular,

$$d_N \circ \varphi - \varphi \circ d_M = 0 = d^0(\varphi).$$

Note that $d_N \circ \varphi : M \rightarrow N[1]$ and $\varphi \circ d_M : M \rightarrow N[1]$, so this equality indeed holds. Moreover, suppose we have any graded R -module homomorphism $\psi : M \rightarrow N$ such that $d_N \circ \psi - \psi \circ d_M = 0$. Then $d_N \circ \psi = \psi \circ d_M$, so ψ is a differential graded map. Thus if we define

$$d^0(\psi) := d_N \circ \psi - \psi \circ d_M$$

for all $\psi \in \underline{\mathcal{C}h}_R(M, N)^0$, then we obtain that the possible choices of $f^0(1)$ are precisely differential graded maps $\mathcal{C}h_R(M, N)$, giving us our identification.

Since there is now a correspondence $\text{Hom}_{R^0}(R^0, \text{Hom}_R(M, N)) \cong \mathcal{C}h_R(M, N)$ of only degree 0 maps with degree 0 homogeneous elements, we should have that there is precisely one way to define the higher degrees of any such map. Consider a differential graded R -module morphism $f : R \rightarrow \underline{\mathcal{C}h}_R(M, N)$. By the graded ring structure explored in discussion 1.4.2, since $1 \in R$ is a unit, we obtain that $r = r \cdot 1$. Since f is a graded R -module homomorphism and 1 is homogeneous of degree 0, we have by discussion 1.4.11 that $f^i(r \cdot 1) = r \cdot f^0(1)$. Thus f^i is uniquely determined by f^0 .

We now want to define d^i fully generally. Consider d^1 . If we defined $d^1(f) = d_N \circ f - f \circ d_M$, then we would not obtain $d^1 \circ d^0 = 0$: let $f : M \rightarrow N$ be a differential graded R -module morphism. Then

$$\begin{aligned} (d^1 \circ d^0)(f) &= d^1(d_N \circ f - f \circ d_M) = d_N \circ d_N \circ f - d_N \circ f \circ d_M - d_N \circ f \circ d_M + f \circ d_M \circ d_M \\ &= -2d_N \circ f \circ d_M \\ &\neq 0. \end{aligned}$$

It appears that there has been a sign mismatch which made cancellation fail. Indeed, this issue is resolved by instead defining $d^1(f) = d_N \circ f + f \circ d_M$. The same logic shows that we must have $d^2(f) = d_N \circ f - f \circ d_M$, or else we would have the same sign mismatch, and this shows $d^3(f) = d_N \circ f + f \circ d_M$, and so on. In full generality we can thus define

$$d^i(f) = d_N \circ f - (-1)^{|f|} f \circ d_M.$$

Notation 2.1.9. From now on, we write the internal hom of $\mathcal{C}h(R)$ as $\underline{\text{Hom}}_R(M, N)$, since it is the same as that of $\mathcal{G}\text{Mod}_R$. However, we continue writing the unenriched hom-sets as $\mathcal{C}h_R(M, N)$.

Construction 2.1.10. We now promote the graded tensor product of definition 1.1.8 and remark 1.4.13 to $\mathcal{C}h(R)$ such that $\mathcal{C}h(R)$ is closed monoidal with internal hom as given in construction 2.1.8. We would also like the evident forgetful functor $U : \mathcal{C}h(R) \rightarrow \mathcal{G}\text{Mod}_R$ to be monoidal, so we are forced to define the tensor in $\mathcal{C}h(R)$ in the same way as in $\mathcal{G}\text{Mod}_R$. All we need to define are the differentials.

Let M, N be differential graded R -modules. Recall the concrete construction of the tensor product

$$\begin{aligned} (M \otimes_R N)^k &= \frac{\bigoplus_{i,j \in \mathbb{Z}}^{i+j=k} M^i \otimes_{\mathbb{Z}} N^j}{\langle rm \otimes n - m \otimes rn \mid r \in R, m \in M, n \in N, |r| + |m| + |n| = k \rangle} \\ &= \frac{\bigoplus_{i,j \in \mathbb{Z}}^{i+j=k} M^i \otimes_{\mathbb{Z}} N^j}{\bigoplus_{p,q,d \in \mathbb{Z}}^{p+q+d=k} \langle R^d M^p \otimes_{\mathbb{Z}} N^q - M^p \otimes_{\mathbb{Z}} R^d N^q \rangle}. \end{aligned}$$

It suffices to construct $\mathbb{Z}$ -linear maps $d^i : (M \otimes_R N)^{i-1} \rightarrow (M \otimes_R N)^i$ which are R -linear in the relevant sense. We will do this by enforcing the tensor-hom adjunction.

As explored in ??, the adjunct $\text{Ch}_R(M \otimes_R N, P) \cong \text{Ch}_R(M, \widehat{\text{Hom}}_R(N, P))$ is given by mapping $f : M \otimes_R N \rightarrow P$ to a map $\widehat{f} : M \rightarrow \widehat{\text{Hom}}_R(N, P)$ given by $\widehat{f}(m) = f(m \otimes -)$ for any homogeneous element $m \in M$. To simplify notation, write d_h for the differential on the internal hom. For the tensor-hom adjunction to hold in Ch_R , this map must commute with differentials in the sense that

$$d_h \circ \widehat{f} = \widehat{f} \circ d_M.$$

We evaluate both sides on a homogeneous element $m \in M$, yielding the equality

$$d_h(\widehat{f}(m)) = \widehat{f}(d_M(m)).$$

Note that $|\widehat{f}(m)| = |m|$ since $\widehat{f}$ is a morphism of graded modules. Applying this to the formula for d_h found in construction 2.1.8, we have that

$$d_h(\widehat{f}(m)) = d_P \circ \widehat{f}(m) - (-1)^{|m|} \widehat{f}(m) \circ d_M.$$

We now evaluate both sides on a homogeneous element $n \in N$:

$$d_P(f(m \otimes n)) - (-1)^{|m|} f(m \otimes d_N(n)) = f(d_M(m) \otimes n).$$

Rearranging yields

$$d_P(f(m \otimes n)) = f(d_M(m) \otimes n) + (-1)^{|m|} f(m \otimes d_N(n)) = f(d_M(m) \otimes n + (-1)^{|m|} m \otimes d_N(n))$$

by $\mathbb{Z}$ -linearity of f . Since this must hold on all $m \otimes n \in M \otimes_R N$, we obtain that

$$d_P \circ f = f \circ (d_M \otimes_R 1_N + (-1)^{|\cdot|} 1_M \otimes_R d_N).$$

Since f is a chain map, we have $d_P \circ f = f \circ d_{M \otimes_R N}$. We can choose f, P such that f is monic. Then we obtain

$$d_{M \otimes_R N}(m \otimes n) = d_M(m) \otimes n + (-1)^{|m|} m \otimes d_N(n).$$

This defines the differential on $M \otimes_R N$. Note that it is R -bilinear in the relevant sense due to R -linearity of d_M and d_N .

Corollary 2.1.11. *The inclusion functor $\mathcal{G}\text{Mod}_R \rightarrow \text{Ch}_R$ which views graded R -modules as having 0 differential is monoidal. The forgetful functor $U : \text{Ch}_R \rightarrow \mathcal{G}\text{Mod}_R$ is monoidal.*

2.2 Differential graded associative and commutative R -algebras

Definition 2.2.1. A differential graded associative R -algebra is an algebra for the operad $\mathcal{A}\text{ss}$ in $\text{Ch}(R)$. The category of differential graded R -algebras is denoted $\mathcal{D}\mathcal{G}\text{Alg}_R$.

Remark 2.2.2. Let $\mathcal{O}$ be a (symmetric) colored operad and $F : \mathcal{O} \rightarrow \mathcal{C}$ a (symmetric) algebra for $\mathcal{O}$ over a symmetric monoidal category $\mathcal{C}$. Let $U : \mathcal{C} \rightarrow \mathcal{D}$ be a monoidal functor between symmetric monoidal categories. Then the composite $U \circ F : \mathcal{O} \rightarrow \mathcal{D}$ is a (symmetric) algebra for $\mathcal{O}$ over $\mathcal{D}$. In particular, construction 2.1.8 dictates that if A is a differential graded associative R -algebra, then composing with the functor $\text{Ch}(R) \rightarrow \mathcal{G}\text{Mod}_R$ yields a graded associative R -algebra. Since this functor simply forgets the differentials, we obtain that a differential graded associative R -algebra is precisely a graded associative R -algebra together with some differentials.

Discussion 2.2.3. We will now show that the differential for a differential graded R -algebra (A, d_A) is a *derivation*, meaning it satisfies a Leibniz rule $d_A(xy) = d_A(x)y + (-1)^{|x|}xd_A(y)$ for $x, y \in A$ homogeneous elements. To do this, we use the fact that since the multiplication map $\mu : A \otimes_R A \rightarrow A$ is a morphism in $\mathbf{Ch}_R$, it must satisfy $\mu \circ d_{A \otimes_R A} = d_A \circ \mu$. We have

$$\begin{aligned} (d \circ \mu)(x \otimes y) &= d(xy) = \mu \left(d_A(x) \otimes y + (-1)^{|x|}x \otimes d_A(y) \right) = \mu(d_A(x) \otimes y) + (-1)^{|x|}\mu(x \otimes d_A(y)) \\ &= d_A(x)y + (-1)^{|x|}xd_A(y). \end{aligned}$$

Definition 2.2.4. A differential graded commutative R -algebra is a commutative algebra for the operad $\mathbf{Com}$ in $\mathbf{Ch}(R)$. The category of differential graded commutative R -algebras is denoted $\mathbf{DGCAlg}_R$.

Discussion 2.2.5. All of the properties of differential graded commutative algebras have already been explored in discussion 2.2.3 and discussion 1.3.8: they are graded R -algebras satisfying graded commutativity in the sense that $xy = (-1)^{|x||y|}yx$, and have a differential satisfying the Leibniz rule $d(xy) = d(x)y + (-1)^{|x|}xd(y)$ for all homogeneous elements x, y .

An example of great importance is the graded symmetric R -algebra on a graded R -module. We begin with a review of linear algebra.

Discussion 2.2.6. For now, let R be a normal ring and M an R -module. Recall that the symmetric algebra $\mathbf{Sym}^\bullet(M)$ is given by the quotient of the tensor algebra $T(M)$ by the subalgebra generated by elements of the form $x \otimes y - y \otimes x$. We then denote equivalence classes $[x \otimes y]$ by xy , and the quotient forces $xy = yx$. We would like to recharacterize this construction in a more general manner.

We can start by recharacterizing the tensor algebra. Given an R -module M , the tensor algebra $T(M)$ can intuitively be thought of as the free associative R -algebra on M . The way to formalize such a notion is to say that the tensor algebra defines a functor $T : \mathbf{Mod}_R \rightarrow \mathbf{Alg}_R$ which is left adjoint to the evident forgetful functor $U : \mathbf{Alg}_R \rightarrow \mathbf{Mod}_R$. It is thus natural to think of the symmetric algebra as defining a functor $\mathbf{Sym}^\bullet : \mathbf{Mod}_R \rightarrow \mathbf{CAlg}_R$ which is left adjoint to the forgetful functor $U : \mathbf{CAlg}_R \rightarrow \mathbf{Mod}_R$; hence $\mathbf{Sym}^\bullet$ is the free commutative algebra on a module. This suggests the following definition.

Definition 2.2.7. Let M be a graded R -module. Then the *graded symmetric R -algebra on M* , denoted $\mathbf{Sym}^\bullet(M)$, is a commutative algebra which admits a map $i : M \rightarrow \mathbf{Sym}^\bullet(M)$ which is universal in the sense that precomposition induces isomorphisms

$$i^* : \mathbf{CAlg}_R(\mathbf{Sym}^\bullet(M), A) \cong \mathbf{Hom}_R(M, U(A))$$

natural in A , where $U : \mathbf{CAlg}_R \rightarrow \mathbf{GMod}_R$ is the forgetful functor.

Discussion 2.2.8. Given a graded R -module M , we can define a tensor algebra $T(M)$ by taking $T(M) := \bigoplus_{i \geq 0} M^{\otimes i}$, where the tensor is the graded tensor product taken over R . Here we define the 0th tensor power to be R . Although we will not show this, it will not be unexpected that T is universal in the sense a tensor algebra must be universal. We might expect that $\mathbf{Sym}^\bullet(M)$ can be derived by applying the proper quotient to $T(M)$ such that the algebra $\mathbf{Sym}^\bullet(M) : \mathbf{Ass} \rightarrow \mathbf{GMod}_R$ becomes a commutative algebra $\mathbf{Com} \rightarrow \mathbf{GMod}_R$. As explored in discussion 1.3.8, this amounts to quotienting by the S_n -action on $T(M)$.

We explore once again what that means. Let $\alpha : \langle 2 \rangle \rightarrow \langle 2 \rangle$ be the permutation $1 \mapsto 2 \mapsto 1$. This acts on symmetric monoidal categories by precomposing by the symmetry isomorphism,

which we will write as σ^* . Since this acts trivially on S_n , we must identify the multiplication $\mu : T(M) \otimes_R T(M) \rightarrow T(M)$ with $\sigma^* \mu : T(M) \otimes_R T(M) \rightarrow T(M)$. Since symmetry is given by Koszul braiding, this implies that we have an equality between $\mu(x \otimes y) = xy$ and $(\mu \circ \sigma)(x \otimes y) = (-1)^{|x||y|}yx$. Of course since α generates the S_n action on symmetric monoidal categories, we obtain that

$$\text{Sym}(M) = \frac{T(M)}{\langle xy - (-1)^{|x||y|}yx \mid x, y \in T(M) \rangle}.$$

Recall here that the quotient is taken degreewise.

We now show that this definition of $\text{Sym}^\bullet(M)$ satisfies definition 2.2.7. Define the map $i : M \rightarrow \text{Sym}^\bullet(M)$ as follows. Since $M = M^{\otimes 1}$, there is a natural inclusion $M \hookrightarrow T(M)$. Postcomposing by the projection $T(M) \rightarrow \text{Sym}^\bullet(M)$ yields a map $i : M \rightarrow \text{Sym}(M)$. Since i^* yields a morphism $\mathcal{C}\text{Alg}_R(\text{Sym}^\bullet(M), A) \rightarrow \text{Hom}_R(M, U(A))$, we simply show that a morphism $f : M \rightarrow U(A)$ induces a morphism $\tilde{f} : \text{Sym}^\bullet(M) \rightarrow A$ with $\tilde{f} \circ i = f$.

The map i is trivially shown to be pointwise injective, so we abuse notation and write $i(x) = x$ for $x \in M$ a homogeneous element. Let $f : M \rightarrow U(A)$ be a morphism of graded R -modules. For $\tilde{f}$ to be a morphism of graded R -algebras, we must have $\tilde{f}(x_1 \cdots x_n) = \tilde{f}(x_1) \cdots \tilde{f}(x_n)$. We then define $\tilde{f}(x_i) = f(x_i)$. This manifestly satisfies $\tilde{f} \circ i = f$, yielding our adjunction.

Example 2.2.9. Let R be a regular ring and M be a regular R -module concentrated in degree 0, such that $M[-1]$ is concentrated in degree 1. Then $\text{Sym}^\bullet(M) \cong \Lambda^\bullet M$, where $(\Lambda^\bullet M)^i = \Lambda^i M$ is the i th exterior power of M over R .

Definition 2.2.10. Let M be a differential graded R -module. Then the *differential graded symmetric R -algebra* on M , denoted $\text{Sym}^\bullet(M)$, is the differential graded commutative algebra which is universal in the sense that

$$\mathcal{D}\mathcal{G}\mathcal{C}\text{Alg}_R(\text{Sym}^\bullet(M), A) \cong \mathcal{C}\text{h}_R(M, U(A))$$

natural in A , where $U : \mathcal{D}\mathcal{G}\mathcal{C}\text{Alg}_R \rightarrow \mathcal{C}\text{h}_R$ is the forgetful functor.

Construction 2.2.11. Let (M, d_M) be a differential graded R -module. We define a differential d_S on the graded symmetric algebra $\text{Sym}^\bullet(M)$ to construct the differential graded symmetric algebra; the same logic as discussion 2.2.8 proves that it solves the universal problem in the differential graded categories. We need the inclusion $i : M \rightarrow \text{Sym}^\bullet(M)$ to be a morphism of differential graded R -modules, meaning $d_S \circ i = i \circ d_M$. If we view i as restriction along M , then this implies that $d_S|_M = d_M$. By discussion 2.2.3, d_S must satisfy the graded Leibniz rule. This forces

$$d_S(x) = d_M(x), \quad d_S(xy) = d_M(x)y + (-1)^{|x|}d_M(y)$$

for all homogeneous elements $x, y \in M$. Inducting on this definition shows

$$d_M(x_1 \cdots x_n) = \sum_{i=1}^n (-1)^{\sum_{j=1}^{i-1} |x_j|} x_1 \cdots d_M(x_i) \cdots x_n.$$

A mildly annoying proof shows that this derivation is indeed an R -linear map of differential graded R -modules, completing the proof.

2.3 Differential graded Lie R -algebras

We now want to define differential graded Lie R -algebras over a graded ring R . Doing this intuitively is nontrivial, since the Lie operad is not particularly simple to define. We start by assuming R is a normal ring.

Definition 2.3.1. Let $(\mathcal{V}, \otimes, I)$ be a monoidal category, although usually it is a closed symmetric monoidal category. A $\mathcal{V}$ -enriched colored operad $\mathcal{O}$, or simply a colored $\mathcal{V}$ -operad, is

- A collection $\text{Obj}(\mathcal{O})$ of *objects* or *colors*.
- For any $n \geq 0$, any $\{x_i\}_{1 \leq i \leq n} \subseteq \text{Obj}(\mathcal{O})$, and any $y \in \text{Obj}(\mathcal{O})$, an object $\mathcal{O}(\{x_i\}, y)$ of $\mathcal{V}$ called a *hom-object* of *n-multimorphisms*.
- For any $x \in \text{Obj}(\mathcal{O})$, a morphism $1_x : I \rightarrow \mathcal{O}(x, x)$ called the *identity morphism on x*.
- For any $n \geq 1$, any objects $\{x_i\}_{1 \leq i \leq n} \subseteq \text{Obj}(\mathcal{O})$, integers $k_1, \dots, k_n$, and sequences of objects $\{w_{i,j}\}_{1 \leq j \leq k_i}$ for all $1 \leq i \leq n$, a composition map

$$\mathcal{O}(\{x_i\}, y) \otimes \mathcal{O}(\{w_{1,j}\}, x_1) \otimes \dots \otimes \mathcal{O}(\{w_{n,j}\}, x_n) \rightarrow \mathcal{O}(\{w_{i,j}\}_{1 \leq i \leq n, 1 \leq j \leq k_i}, y).$$

- This data must satisfy associativity and unit laws, as expected.

The operad is called *symmetric* if composition is S_n -equivariant.

Definition 2.3.2. Fix a ring R . The operad $\mathfrak{Lie}_R$ is the symmetric colored Mod_R -operad with one color. For simplicity of notation, write $\mathfrak{Lie}_R(n)$ for the set of n -multimorphisms $\mathfrak{Lie}_R(\{*, \dots, *\}, *)$.

- Let V_n be a free R -module on n letters. Let $F(V_n)$ be the *free Lie R -algebra* on V_n , formally defined as the left adjoint to the forgetful functor. To construct $F(V_n)$ explicitly, let $T(V_n)$ be the tensor R -algebra, which forms a Lie R -algebra via $[x, y] := x \otimes y - y \otimes x$. Then $F(V_n)$ is the Lie subalgebra generated by $V_n \subseteq T(V_n)$.
- An element of $F(V_n)$ is thus a nested collection of commutators of elements of V_n . For $x_1, \dots, x_n \in V_n$, define $P(x_1, \dots, x_n)$ to be any nested commutator of n elements, called a *Lie polynomial of degree n*. For example, $[x_1, [x_2, x_3]]$ and $[[x_1, x_2], x_3]$ are both valid nested commutators of 3 elements.
- Define the space $\mathfrak{Lie}_R(n)$ of n -multimorphisms as the subspace of $F(V_n)$ spanned by elements of the form $P(x_1, \dots, x_n)$ where each x_i appears exactly once.
- Given $P(x_1, \dots, x_n) \in \mathfrak{Lie}_R(n)$ and $\sigma \in S_n$, define the S_n -action to be

$$(P \cdot \sigma)(x_1, \dots, x_n) = P(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

- Let $P \in \mathfrak{Lie}_R(m)$, let $f : \langle n \rangle \rightarrow \langle m \rangle$, and let $Q_j \in \mathfrak{Lie}_R(f^{-1}(j))$ for all $j \in \langle m \rangle$. The composition map is defined as $Q_1 \otimes_R \dots \otimes_R Q_m \otimes_R P \mapsto P(Q_1, \dots, Q_m)$, where the notation means substitution. For example, if $P(x_1, x_2) = [x_1, x_2]$, if $Q_1(y_1, y_2, y_3) = [[y_1, y_2], y_3]$, and $Q_2(z_1, z_2, z_3) = [z_1, [z_2, z_3]]$, then

$$P(Q_1, Q_2) = [[[y_1, y_2], y_3], [z_1, [z_2, z_3]]].$$

Discussion 2.3.3. The definition of the $\mathfrak{Lie}_R$ operad is highly involved and deserves significant analysis. The reason for requiring enrichment over Mod_R is twofold. First, the Jacobi identity requires enrichment over abelian groups in some sense, since we need to be able to add and negate morphisms. Additionally, R -multilinearity of the Lie bracket requires R -enrichment. Thus we will see later that a Lie R -algebra is an algebra over the operad $\mathfrak{Lie}_R$ in Mod_R . For now, we investigate the structure of construction 3.2.6. We start with the basic structure of the spaces of multimorphisms.

Since $V_0 \cong 0$, we see that $F(V_0) \cong 0$ contains exactly one element. One can define the zero Lie polynomial $P(0) = 0$, showing that $\mathfrak{Lie}_R(0) \cong 0$. As seen in discussion 1.2.7, this will guarantee that the element 0 of a Lie R -algebra annihilates higher arity operations.

Since V_1 is 1-dimensional, any nonzero element forms a basis. Let $x \in V_1$ be nonzero. The only valid Lie polynomial of degree 1 is $P(x) = x$, so there is a unique morphism $* \rightarrow *$ in $\mathfrak{Lie}_R(1)$. Define this morphism to be the identity element, meaning we have $\mathfrak{Lie}_R(1) \cong \langle P = 1_x \rangle$ is the free R -module on the identity morphism. Here it's useful to recall that polynomials themselves need not form a R -module; for example $P + P = 2P \neq P$, so the set $\{P\}$ is not a R -module. Thus we must generate a space on P . The reason that $\mathfrak{Lie}_R(0) \cong 0$ was since the zero polynomial satisfied $k0 = 0$.

Now we consider V_2 , which is where we expect to see some notion of a Lie bracket. Since V_2 has dimension 2, we can choose a basis $\{x, y\}$ for V_2 . The only valid Lie polynomials of degree 2 are $P(x, y) = [x, y]$ and $Q(x, y) = [y, x]$. However, since the Lie bracket is taken in the tensor R -algebra $T(V_2)$ which is itself a Lie R -algebra, we have the identity $[x, y] = -[y, x]$. Thus we write $Q = -P$, and find that $\mathfrak{Lie}_R(2) \cong \langle P \rangle$ is the 1-dimensional free R -module generated by $P(x, y) = [x, y]$. It will be important to note for discussion 2.3.9 that if $\tau = (12) \in S_2$ is the nontrivial permutation, then $P \cdot \tau = Q = -P$.

Now consider V_3 , which will be the final space we consider. We expect at this point to see the Jacobi identity, which is the only axiom of a Lie R -algebra which requires arity 3, so we recall what the Jacobi identity should say. If $x, y, z \in \mathfrak{g}$ are elements of some Lie R -algebra $\mathfrak{g}$, then

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

Write $\mu(x, y) = [x, y]$ as the 2-multimorphism which should encode the Lie bracket. Let $x_1, x_2, x_3 \in V_3$ be generators. It is clear that $\mathfrak{Lie}_R$ is spanned by all permutations of the two possible Lie bracket combinations $P(x_1, x_2, x_3) = [x_1, [x_2, x_3]]$ and $Q(x_1, x_2, x_3) = [[x_1, x_2], x_3]$. However, by antisymmetry of the Lie bracket,

$$Q(x_1, x_2, x_3) = [[x_1, x_2], x_3] = -[x_3, [x_1, x_2]] = P(x_3, x_1, x_2).$$

Thus Q is in the span of the permutations of P . We find that $\mathfrak{Lie}_R(3) \cong \langle P \cdot S_3 \rangle$. We also know that, since $T(V_3)$ is a Lie R -algebra, the Jacobi identity implies

$$P(x_1, x_2, x_3) + P(x_2, x_3, x_1) + P(x_3, x_1, x_2) = 0.$$

However, the Jacobi identity should be written in terms of the Lie bracket μ , not the Lie polynomial P . Thus we want to rewrite this as

$$(\mu \circ (1_*, \mu))(x_1, x_2, x_3) + (\mu \circ (1_*, \mu))(x_2, x_3, x_1) + (\mu \circ (1_*, \mu))(x_3, x_1, x_2) = 0.$$

Here 1_* is the identity morphism. It thus suffices to show $P = \mu \circ (1_*, \mu)$. Recall from construction 3.2.6 that composition is given by substitution, meaning $\mu \circ (1_*, \mu) = \mu(1_*, \mu)$. Evaluating on the basis yields

$$(\mu(1_*, \mu))(x_1, x_2, x_3) = \mu(1_*(x_1), \mu(x_2, x_3)) = [x_1, [x_2, x_3]] = P(x_1, x_2, x_3).$$

Thus the Jacobi identity holds.

Definition 2.3.4. Let R be a ring. A *Lie R -algebra* is an algebra for $\mathfrak{Lie}_R$ over Mod_R . The category of Lie R -algebras, denoted Lie_R , is the category $\mathcal{CAlg}_{\mathfrak{Lie}_R}(\text{Mod}_R)$.

Discussion 2.3.5. The structure of Lie R -algebras should already be fairly well-understood from definition 2.3.4, especially since the category of enrichment is concrete, meaning the study of enriched category theory is simplified by identifying with the underlying category. Let $\mathfrak{g}$ be a Lie R -algebra. All arity $n \geq 2$ operations are generated by the Lie bracket μ and the identity morphism. The Jacobi

identity and antisymmetry of μ are satisfied. There is a distinguished element $0 : R \rightarrow \mathfrak{g}$ which annihilates μ in the relevant sense. Morphisms of Lie R -algebras $\mathfrak{g} \rightarrow \mathfrak{h}$ are morphisms of R -modules which preserve μ in the relevant sense. We are more interested in how to promote this to the graded setting. Since we required $\mathfrak{L}ie_R$ to be enriched over $\mathcal{M}od_R$ in order to define Lie R -algebras, we might expect a graded or differential graded Lie operad to be enriched over graded or differential graded R -modules. This would also allow for R to be a graded ring. We do this by concentrating the operad of construction 3.2.6 in degree 0, with trivial differential.

Definition 2.3.6. Let R be a graded ring. A *free graded R -module* $F(S)$ is defined on a collection $S = \{S_i\}_{i \in \mathbb{Z}}$ of sets, and is defined in the way one would expect: $F(S)^k$ consists of all formal linear combinations of formal elements rs , where $r \in R$ and $s \in S$ are homogeneous elements with $|r| + |s| = k$. If S is concentrated in degree 0, then so is $F(S)$, and we refer to $F(S)$ as a *free R -module*.

Definition 2.3.7. Let R be a graded ring.

- (1) We can define the operad $\mathfrak{g}\mathfrak{L}ie_R$ in the same way as construction 3.2.6 by taking the generators $\{x_1, \dots, x_n\}$ of V_n to be concentrated in degree 0.
- (2) The operad $\mathfrak{d}\mathfrak{g}\mathfrak{L}ie_R$ is defined by giving $\mathfrak{g}\mathfrak{L}ie_R$ the 0 differential pointwise.

In particular if R is concentrated in degree 0, then $\mathfrak{g}\mathfrak{L}ie_R$ and $\mathfrak{d}\mathfrak{g}\mathfrak{L}ie_R$ are simply the operad $\mathfrak{L}ie_R$ concentrated in degree 0 pointwise.

Definition 2.3.8. Let R be a graded ring.

- (1) A *graded Lie R -algebra* is an algebra for $\mathfrak{g}\mathfrak{L}ie_R$ over $\mathcal{G}Mod_R$. The category of graded Lie R -algebras, denoted $\mathcal{G}Lie_R$, is the category $\mathcal{C}Alg_{\mathfrak{g}\mathfrak{L}ie_R}(\mathcal{G}Mod_R)$.
- (2) A *differential graded Lie R -algebra* is an algebra for $\mathfrak{d}\mathfrak{g}\mathfrak{L}ie_R$ over $\mathcal{C}h_R$. The category of differential graded Lie R -algebras, denoted $\mathcal{D}GLie_R$, is the category $\mathcal{C}Alg_{\mathfrak{d}\mathfrak{g}\mathfrak{L}ie_R}(\mathcal{C}h_R)$.

Discussion 2.3.9. We discuss graded Lie R -algebras first. Let $\mathfrak{g}$ be a graded Lie R -algebra. We see first that $\mathfrak{g}$ is a graded R -module. By proposition 3.2.7, we see that there is a map $[-, -] : \mathfrak{g} \otimes_R \mathfrak{g} \rightarrow \mathfrak{g}$ induced by the 2-ary generator $\mu \in \mathfrak{g}\mathfrak{L}ie_R(2)^0$. Let $x, y \in \mathfrak{g}$ be homogeneous elements. We will start by showing that $[-, -] = \mathfrak{g}(\mu)$ satisfies *graded antisymmetry*, meaning $[x, y] = -(-1)^{|x||y|}[y, x]$.

It will be useful to view $\mathfrak{g}$ as a functor of symmetric colored $\mathcal{G}Mod_R$ -operads, meaning $[-, -] = \mathfrak{g}(\mu)$. Since $\mathfrak{g}$ is a $\mathcal{G}Mod_R$ -functor, it preserves scalar multiplication. Since it is a symmetric functor, it preserves the S_n -action. Recall from example 1.3.4 that the induced action by τ on the symmetric monoidal category $\mathcal{G}Mod_R$ is precomposition by the symmetry isomorphism σ given by Koszul braiding. Finally, recall from proposition 3.2.7 that $\mu \cdot \tau = -\mu$. Thus

$$[-, -] \circ \sigma = \mathfrak{g}(\mu) \circ \sigma = \mathfrak{g}(\mu \cdot \tau) = \mathfrak{g}(-\mu) = -\mathfrak{g}(\mu) = -[-, -].$$

This implies

$$(-1)^{|x||y|}[y, x] = ([-, -] \circ \sigma)(x \otimes y) = (-[-, -])(x \otimes y) = -[x, y].$$

Multiplying both sides by -1 yields

$$[x, y] = -(-1)^{|x||y|}[y, x].$$

Therefore, Lie brackets for graded Lie R -algebras satisfy a graded antisymmetry rule.

We also remark that since $[-, -] : \mathfrak{g} \otimes_R \mathfrak{g} \rightarrow \mathfrak{g}$ is a morphism of graded R -modules, it must preserve degrees. Using the explicit construction of the graded tensor product in construction 1.4.14, it follows that $[x, y]$ has degree $|x \otimes y| = |x| + |y|$. This is usually written as $[\mathfrak{g}^i, \mathfrak{g}^j] = \mathfrak{g}^{i+j}$.

We next show the Lie bracket satisfies a *graded Jacobi identity*, which is usually written as

$$[x, [y, z]] = [[x, y], z] + (-1)^{|x||y|}[y, [x, z]]. \quad (*)$$

The Jacobi identity in $\mathfrak{g}\mathfrak{Lie}_R$ can be rewritten using the permutations $\tau_1 = (123)$ and $\tau_2 = (132) = (123)^2$. We have

$$(\mu \circ (1, \mu)) + (\mu \circ (1, \mu)) \cdot \tau_1 + (\mu \circ (1, \mu)) \cdot \tau_2 = 0.$$

Mac Lane's coherence axioms for symmetric monoidal categories prove that we may write the action of these permutations on $\mathcal{G}\text{Mod}_R$ as follows:

$$\begin{array}{ccccc} & & \tau_1 & & \\ & \nearrow & & \searrow & \\ x \otimes y \otimes z & \xrightarrow{\sigma \otimes 1} & y \otimes x \otimes z & \xrightarrow{1 \otimes \sigma} & y \otimes z \otimes x, \\ & \nwarrow & & \nearrow & \\ x \otimes y \otimes z & \xrightarrow{1 \otimes \sigma} & x \otimes z \otimes y & \xrightarrow{\sigma \otimes 1} & z \otimes x \otimes y. \\ & & \tau_2 & & \end{array}$$

Since $\mathfrak{g}$ is a $\mathcal{G}\text{Mod}_R$ -functor and a functor of symmetric colored operads, it preserves the S_3 -action, addition, and the zero element. We thus have

$$\begin{aligned} \mathfrak{g}((\mu \circ (1, \mu)) + (\mu \circ (1, \mu)) \cdot \tau_1 + (\mu \circ (1, \mu)) \cdot \tau_2) &= \mathfrak{g}(\mu \circ (1, \mu)) + \mathfrak{g}(\mu \circ (1, \mu)) \circ \tau_1 + \mathfrak{g}(\mu \circ (1, \mu)) \circ \tau_2 \\ &= [-, [-, -]] + [-, [-, -]] \circ \tau_1 + [-, [-, -]] \circ \tau_2 \\ &= 0. \end{aligned}$$

We evaluate this on an element $x \otimes y \otimes z$:

$$\begin{aligned} &([-, [-, -]] + [-, [-, -]] \circ \tau_1 + [-, [-, -]] \circ \tau_2)(x \otimes y \otimes z) \\ &= [x, [y, z]] + (-1)^{|x||y|}(-1)^{|x||z|}[y, [z, x]] + (-1)^{|y||z|}(-1)^{|x||z|}[z, [x, y]] = 0. \end{aligned} \quad (**)$$

One form of the Jacobi identity (the one listed on Wikipedia) is derived from this as follows. We multiply both sides by $(-1)^{|x||z|}$ to obtain

$$(-1)^{|x||z|}[x, [y, z]] + (-1)^{|x||y|}[y, [z, x]] + (-1)^{|y||z|}[z, [x, y]] = 0.$$

The other form of the Jacobi identity $(*)$ is derived from $(**)$ as follows.

We can rewrite the second and third terms using graded antisymmetry. For the second term, applying graded antisymmetry to the inner bracket yields $[z, x] = -(-1)^{|x||z|}[x, z]$, so

$$(-1)^{|x||y|+|x||z|}[y, [z, x]] = -(-1)^{|x||y|+2|x||z|}[y, [x, z]] = -(-1)^{|x||y|}[y, [x, z]].$$

For the third term, recall that the degree of $[x, y]$ is $|x| + |y|$. Applying graded antisymmetry to the outer bracket yields $[z, [x, y]] = -(-1)^{|z|(|x|+|y|)}[[x, y], z]$, so

$$(-1)^{|y||z|+|x||z|}[z, [x, y]] = -(-1)^{2|y||z|+2|x||z|}[[x, y], z] = -[[x, y], z].$$

Substituting these back into our evaluated sum gives

$$[x, [y, z]] - (-1)^{|x||y|}[y, [x, z]] - [[x, y], z] = 0,$$

which rearranging yields equation $(*)$:

$$[x, [y, z]] = [[x, y], z] + (-1)^{|x||y|}[y, [x, z]].$$

Thus we find that a graded Lie R -algebra is simply a graded R -module $\mathfrak{g}$ with a graded antisymmetric map $[-, -] : \mathfrak{g} \otimes_R \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying a graded Jacobi identity.

Discussion 2.3.10. We now discuss differential graded Lie R -algebras. Let $\mathfrak{g}$ be a differential graded Lie R -algebra. Since this is an object of $\mathcal{Ch}_R$ (with some extra algebraic properties), it has a differential d . By corollary 2.1.11, the forgetful functor $U : \mathcal{Ch}_R \rightarrow \mathcal{GMod}_R$ is monoidal, so if $\mathfrak{g}$ is a differential graded Lie R -algebra, then $U \circ \mathfrak{g}$ is a graded Lie R -algebra. Thus the underlying graded structure of $\mathfrak{g}$ is described by discussion 2.3.9, meaning we only need to consider the structure of the differential d .

We will show that d acts as a derivation on the Lie bracket, meaning for any homogeneous elements $x, y \in \mathfrak{g}$, it satisfies a graded Leibniz rule:

$$d[x, y] = [d(x), y] + (-1)^{|x|}[x, d(y)].$$

Since $\mathfrak{g}$ is an algebra in $\mathcal{Ch}_R$, we find that the Lie bracket must be a map of differential graded R -modules, meaning it commutes with d in the sense that $d \circ [-, -] = [-, -] \circ d_{\mathfrak{g} \otimes \mathfrak{g}}$, where $d_{\mathfrak{g} \otimes \mathfrak{g}}$ is the differential on the tensor product $\mathfrak{g} \otimes \mathfrak{g}$. Recall by construction 2.1.10 that for any $x, y \in \mathfrak{g}$,

$$d_{\mathfrak{g} \otimes \mathfrak{g}}(x \otimes y) = d(x) \otimes y + (-1)^{|x|}x \otimes d(y).$$

We therefore obtain equalities

$$d[x, y] = (d \circ [-, -])(x \otimes y) = ([-, -] \circ d_{\mathfrak{g} \otimes \mathfrak{g}})(x \otimes y) = [-, -] \left(d(x) \otimes y + (-1)^{|x|}x \otimes d(y) \right).$$

By R -linearity of the bracket, this implies

$$d[x, y] = [d(x), y] + (-1)^{|x|}[x, d(y)].$$

Thus, a differential graded Lie R -algebra is simply a graded Lie R -algebra together with a differential which acts as a derivation on the Lie bracket.

2.4 Closure properties

One important thing we have not discussed is the various closure properties of these (differential) graded algebraic structures under taking specific (co)limits and universal constructions. It will be useful to recall the following results.

Proposition 2.4.1 ([ML98] §5.2 Corollary 2). *If a category $\mathcal{C}$ has all small products and equalizers of all pairs of arrows, then $\mathcal{C}$ is complete. Dually, if $\mathcal{C}$ has all small coproducts and all coequalizers of pairs of arrows, then $\mathcal{C}$ is cocomplete.*

Proposition 2.4.2 ([ML98] §5.3 Theorem 1). *Let $\mathcal{J}, \mathcal{C}, \mathcal{D}$ be categories, and let $F : \mathcal{J} \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$ be a functor. If for every $c \in \mathcal{C}$, the functor $\mathcal{J} \rightarrow \mathcal{D}$ given by $F(c)$ has a limit, then F has a limit given by $(\lim F)(c) = \lim(F(c))$. We usually phrase this as limits in functor categories being computed pointwise.*

This has an immediate consequence, which we previously noted in remark 1.1.2.

Proposition 2.4.3. *The category $\mathcal{GMod}_R$ is complete and cocomplete if R is concentrated in degree 0.*

Proof. A graded R -module in this case is a functor $\mathbb{Z} \rightarrow \text{Mod}_R$. Since Mod_R is complete and cocomplete, the statement follows by proposition 2.4.2. ■

Proposition 2.4.4. *The category $\mathcal{GMod}_R$ is complete and cocomplete if R is graded.*

Proof. I don't believe this can be conveniently thought of as a functor category, since we formally defined it as a pullback of a category of functors of colored operads, so we appeal to proposition 2.4.1. The constructions of (co)products and (co)equalizers are pointwise and are the same as in Mod_R . ■

Definition 2.4.5. Let $\mathcal{O}$ be an operad with one color $*$ $\in \mathcal{O}$, and $\mathcal{V}$ a symmetric closed monoidal category. Then the forgetful functor $U : \text{Alg}_{\mathcal{O}}(\mathcal{V}) \rightarrow \mathcal{V}$ is defined on objects by $F \mapsto F(*)$ and morphisms by $\alpha \mapsto (\alpha_* : F(*) \rightarrow G(*))$.

Proposition 2.4.6. Let $\mathcal{O}$ be an operad with one color $*$ $\in \mathcal{O}$, and $\mathcal{V}$ a symmetric closed monoidal category. Then the forgetful functor $U : \text{Alg}_{\mathcal{O}}(\mathcal{V}) \rightarrow \mathcal{V}$ creates limits.

Proof. Let $F : \mathcal{J} \rightarrow \text{Alg}_{\mathcal{O}}(\mathcal{V})$ be a diagram such that the composite $U \circ F$ has a limit $\lim(U \circ F)$. Recall that for U to create limits, we must have that there is a unique limit $\lim F$ such that $U(\lim F) = \lim(U \circ F)$. Since $\mathcal{O}$ is an operad with one color, we simply need to show there is a unique $\mathcal{O}$ -algebra structure on $(U \circ F)(j)$ for all $j \in \mathcal{J}$. We then show that it is a limit of F . Let $j \in \mathcal{J}$, and let $\{\pi_j : \lim(U \circ F) \rightarrow (U \circ F)(j)\}_{j \in \mathcal{J}}$ be the limiting cone. A lot of bullshit happens but it follows. ■

Corollary 2.4.7. The categories $\mathcal{G}\text{Alg}_R$, $\mathcal{G}\mathcal{C}\text{Alg}_R$, $\mathcal{G}\text{Lie}_R$, $\mathcal{D}\mathcal{G}\text{Alg}_R$, $\mathcal{D}\mathcal{G}\mathcal{C}\text{Alg}_R$, and $\mathcal{D}\mathcal{G}\text{Lie}_R$ are complete.

Colimits also follow somehow.

2.5 Homological grading conventions

Throughout this paper, we have used the cohomological grading convention, meaning indices are written as subscripts, and differentials go $d^i : M^{i-1} \rightarrow M^i$. There is another convention, known as the *homological grading convention*, where indices are written as *subscripts* and differentials go $d_i : M_{i-1} \rightarrow M_i$. We view these as the same category by writing $M^i = M_{-i}$ for all i . Although we can view these as the same objects, it is usually useful to view them as different. A differential graded object where the indices decrease is called a *chain complex*, and a differential graded object where the indices increase is called a *cochain complex*. We note that both homological and cohomological degree always coincide, meaning all of our definitions are invariant under change in notation. Finally, we note that we often write chain complexes as $M_{\bullet}$ and cochain complexes as $M^{\bullet}$ to indicate which type of object we view them as. We write $(M_{\bullet})_i = M_i$ and $(M^{\bullet})^i = M^i$.

2.6 Homology and cohomology

Definition 2.6.1. Let $(M^{\bullet}, d^{\bullet})$ be a cochain complex over a graded ring R . The *cohomology* of $M^{\bullet}$ is the graded R -module $H^{\bullet}(M^{\bullet})^i = \text{Ker } d^i / \text{Im } d^{i-1}$. This is viewed as a cochain complex with 0 differential. Similarly, if $(M_{\bullet}, d_{\bullet})$ is a chain complex, the *homology* is the graded R -module $H_{\bullet}(M_{\bullet})_i = \text{Ker } d_{i-1} / \text{Im } d_i$. This is viewed as a chain complex with 0 differential.

Unless otherwise specified, we always work in the cohomological grading convention.

Terminology 2.6.2. If $M^{\bullet}$ is a cochain complex and $x \in M^k$ is a homogeneous element, then if $d(x) = 0$ we say x is a *cocycle* of degree k . If $x = d(y)$ for some $y \in M^{k-1}$, we say x is a *coboundary* of degree k . We drop the 'co' for chain complexes.

Remark 2.6.3. Let M, N be differential graded R -modules. Recall that if $f \in \underline{\text{Hom}}_R(M, N)$ is a homogeneous element of degree k in the internal hom. Recall that $d(f) = d_N \circ f - (-1)^{|f|} f \circ d_M$.

We claim that f is a k -cocycle if and only if $f : M \rightarrow N[k]$ is a morphism of cochain complexes. It suffices only to check compatibility with differentials. If f is a cocycle, then

$$d_N \circ f = (-1)^{|f|} f \circ d_M.$$

By convention, $d_{N[k]} = (-1)^{|k|} d_N = (-1)^{|f|} d_N$, so this implies that f is a morphism of differential graded R -modules. Thus the cohomology of $\underline{\mathrm{Hom}}_R(M, N)$ is given by $\mathrm{Ch}_R(M, N[\bullet])$ modulo coboundaries.

Definition 2.6.4. Let $f, g : M \rightarrow N$ be morphisms of differential graded R -modules. A *homotopy* h from f to g , written $h : f \Rightarrow g$, is a morphism of chain complexes $h : M \rightarrow N[-1]$ such that $d(h) = f - g$, where d is the differential of the internal hom $\underline{\mathrm{Hom}}_R(M, N)$. More explicitly,

$$f - g = d_N \circ h + h \circ d_M.$$

Clearly homotopy is an equivalence relation, so we say f, g are *homotopic* if there is a homotopy $f \Rightarrow g$.

Remark 2.6.5. In view of definition 2.6.4, we see that $H^k(\underline{\mathrm{Hom}}_R(M, N))$ consists of homotopy classes of $\underline{\mathrm{Hom}}_R(M, N)^k = \mathrm{Hom}_R(M, N[k])$. There are ways of viewing Ch_R as a model category or ∞ -category in which the usual notion of homotopy coincides with definition 2.6.4, and the weak equivalences are *quasi-isomorphisms*: morphisms that lower to isomorphisms in (co)homology. This justifies our construction of the internal hom, and thus the graded tensor product.

Chapter 3

Differential graded Lie $\mathbb{K}$ -algebras

Let $\mathbb{K}$ be a field of characteristic zero. We establish the basic facts of differential graded Lie $\mathbb{K}$ -algebras and a particularly important (co)homology theory thereof, which is computed by the Chevalley-Eilenberg complex.

3.1 Lie $\mathbb{K}$ -algebra (co)homology

Let $\mathfrak{g}$ be a Lie $\mathbb{K}$ -algebra.

Definition 3.1.1. View $\mathfrak{g}$ as a differential graded Lie $\mathbb{K}$ -algebra concentrated in degree 0 with trivial differential. The *Chevalley-Eilenberg chain complex* for $\mathfrak{g}$ is the chain complex $C_\bullet(\mathfrak{g})$ given as a graded $\mathbb{K}$ -vector space by

$$C_\bullet(\mathfrak{g}) = \text{Sym}^\bullet(\mathfrak{g}[1]).$$

The differential is given by the Lie bracket.

The *Chevalley-Eilenberg cochain complex* for $\mathfrak{g}$ is the differential graded $\mathbb{K}$ -vector space $C^\bullet(\mathfrak{g})$ given by

$$C^\bullet(\mathfrak{g}) = \underline{\text{Hom}}_{\mathbb{K}}(C_\bullet(\mathfrak{g}), \mathbb{K}).$$

Discussion 3.1.2. Recall here that we take $C_i(\mathfrak{g}) = \text{Sym}^{-i}(\mathfrak{g}[1])$ to reconcile homological and cohomological grading conventions. By definition 2.1.1, we have that $\mathfrak{g}[1]^i = \mathfrak{g}^{i+1}$. Thus

$$\mathfrak{g}^0 = \mathfrak{g}^{1-1} = \mathfrak{g}[1]^{-1}.$$

Thus $\mathfrak{g}[1]$ is concentrated in degree -1 . As seen in example 2.2.9, the symmetric algebra $\text{Sym}(\mathfrak{g}[1])$ can then be given as the exterior algebra

$$\text{Sym}^{-k}(\mathfrak{g}[1]) = \Lambda^k(\mathfrak{g}).$$

It is thus customary to denote the symmetric product in $C_\bullet(\mathfrak{g})$ by the wedge symbol $\wedge$. In particular, $\text{Sym}^0(\mathfrak{g}) = \mathbb{K}$.

The differential d , called the Chevalley-Eilenberg differential, is defined as follows. If $x, y \in \mathfrak{g}$, then we define $d_1(x \wedge y) := -[x, y]$. We would like the differential to and extend this to all of $\text{Sym}(\mathfrak{g})$ by imposing Koszul braiding, bilinearity, and the graded Leibniz rule $d[x, y] = [dx, y] + (-1)^{|x||y|}[x, dy]$. Explicitly, if $x_1, \dots, x_k \in \mathfrak{g}$ are homogeneous elements of $C_\bullet(\mathfrak{g})$ of degree -1 , then

$$d_k(x_1 \wedge \dots \wedge x_k) = \sum_{1 \leq i < j \leq k} (-1)^{i+j} [x_i, x_j] \wedge x_1 \wedge \dots \wedge \widehat{x}_i \wedge \dots \wedge \widehat{x}_j \wedge \dots \wedge x_k,$$

where $\widehat{x}_i$ denotes omitting that argument.

Note here that $C_\bullet(\mathfrak{g})$ is naturally treated as a dg $\mathbb{K}$ -coalgebra. For most of our applications, it is preferred to work with dg $\mathbb{K}$ -algebras, so we generally prefer the Chevalley-Eilenberg cochain complex, which forms a dg $\mathbb{K}$ -algebra.

Discussion 3.1.3. We claim that $C^\bullet(\mathfrak{g}) \cong \text{Sym}^\bullet(\mathfrak{g}^*[-1])$, where $\mathfrak{g}^*$ denotes its dual. To see this, recall a homogeneous element of degree k is a map $C_k(\mathfrak{g}) \rightarrow \mathbb{K}$, which is a map $\text{Sym}^{-k}(\mathfrak{g}[1]) \rightarrow \mathbb{K}$. Since this is just the exterior product, $\alpha \in C^\bullet(\mathfrak{g})$ is a map $\alpha : \Lambda^k \mathfrak{g} \rightarrow \mathbb{K}$.

Linear algebra will now yield our result. There is a canonical form $\langle -, - \rangle : \Lambda^k \mathfrak{g}^* \otimes \Lambda^k \mathfrak{g} \rightarrow \mathbb{K}$ given by

$$\langle \alpha_1 \wedge \cdots \wedge \alpha_k, x_1 \wedge \cdots \wedge x_k \rangle = \det \begin{pmatrix} \alpha_1(x_1) & \cdots & \alpha_1(x_k) \\ \vdots & \ddots & \vdots \\ \alpha_k(x_1) & \cdots & \alpha_k(x_k) \end{pmatrix} = \det \alpha_i(x_j).$$

Since the determinant is alternating, this map is well-defined. Moreover this map is nondegenerate, and thus yields an isomorphism $(\Lambda^k \mathfrak{g})^* \cong \Lambda^k(\mathfrak{g}^*)$ by mapping $\alpha \mapsto \langle \alpha, - \rangle$. Important examples include 1-forms $\langle \alpha, x \rangle = \alpha(x)$ and 2-forms

$$\langle \alpha \wedge \beta, x \wedge y \rangle = \alpha(x)\beta(y) - \alpha(y)\beta(x).$$

Write $d_\bullet$ for the differential of $C_\bullet(\mathfrak{g})$. Recall from construction 2.1.8 that the differential $d^\bullet$ on $C^\bullet(\mathfrak{g})$ is given by

$$d^\bullet \omega = d_{\mathbb{K}} \circ \omega - (-1)^{|\omega|} \omega \circ d_\bullet.$$

As always, we take $d_{\mathbb{K}} = 0$, so the first component vanishes.

We would like to reconcile this differential with the isomorphism we just found. We start in degree 1. Let $\omega \in \mathfrak{g}^*$ and $x, y \in \Lambda^2 \mathfrak{g}$. The isomorphism acts trivially on $\Lambda^1 \mathfrak{g}^* \cong \mathfrak{g}^*$, so we simply have

$$d^1 \omega(x, y) = \langle d_1^* \omega, x \wedge y \rangle = \langle \omega \circ d_1, x \wedge y \rangle = (\omega \circ d_1)(x \wedge y).$$

Recall that (or set $x = x_1$ and $y = x_2$ and plug into the formula)

$$d_1(x \wedge y) = -[x, y].$$

Thus we have

$$d^1 \omega(x, y) = (\omega \circ d_1)(x \wedge y) = -\omega[x, y].$$

We extend this to higher degrees by recalling $C^\bullet(\mathfrak{g})$ is a dg commutative $\mathbb{K}$ -algebra, and ?? says that $d^\bullet$ must be a derivation. Hence

$$d^2(\alpha \wedge \beta) = d^1(\alpha) \wedge \beta + \alpha \wedge d^1(\beta) \text{ THIS IS WRONG.}$$

Here it is useful to recall that we are denoting the symmetric product by the wedge. Inducting on what we have found yields the following two useful formulas. Let $\omega = \alpha_1 \wedge \cdots \wedge \alpha_k$ be a homogeneous element of degree k , and let $x_1, \dots, x_{k+1} \in \mathfrak{g}$. Then

$$d^k \omega(x_1, \dots, x_{k+1}) = \sum_{1 \leq i < j \leq k+1} (-1)^{i+j} \omega([x_i, x_j], x_1, \dots, \widehat{x}_i, \dots, \widehat{x}_j, \dots, x_{k+1})$$

$$d^k(\alpha_1 \wedge \cdots \wedge \alpha_k) = \sum_{i=1}^k (-1)^{i-1} \alpha_1 \wedge \cdots \wedge d(\alpha_i) \wedge \cdots \wedge \alpha_k.$$

Here, as always, the hat denotes omitting an argument.

Definition 3.1.4. The Lie $\mathbb{K}$ -algebra homology of $\mathfrak{g}$ is defined as the homology of the Chevalley-Eilenberg chains $H_{\bullet}^{\text{Lie}}(\mathfrak{g}) := H_{\bullet}(C_{\bullet}(\mathfrak{g}))$. The Lie $\mathbb{K}$ -algebra cohomology of $\mathfrak{g}$ is defined as the cohomology of the Chevalley-Eilenberg cochains $H_{\text{Lie}}^{\bullet}(\mathfrak{g}) := H^{\bullet}(C^{\bullet}(\mathfrak{g}))$.

Discussion 3.1.5. Another approach to Lie $\mathbb{K}$ -algebra cohomology, which we will discuss later on in context of dg Lie algebras, involves computing the derived functors of $\otimes_{\mathbb{K}} U\mathfrak{g}$, where $U\mathfrak{g}$ is the *universal enveloping algebra* of $\mathfrak{g}$. Such a structure is a $\mathbb{K}$ -algebra, and thus a ring. One computes this by noting that there is a quotient of $U\mathfrak{g}$ -modules $U\mathfrak{g}/\mathfrak{g} \cong \mathbb{K}$, and thus $\mathbb{K}$ is a $U\mathfrak{g}$ -module. One then forms the free resolution

$$\cdots \longrightarrow \Lambda^2 \mathfrak{g} \otimes_{\mathbb{K}} U\mathfrak{g} \longrightarrow \Lambda^1 \mathfrak{g} \otimes_{\mathbb{K}} U\mathfrak{g} \longrightarrow \Lambda^0 \mathfrak{g} \otimes_{\mathbb{K}} U\mathfrak{g} \longrightarrow \mathbb{K}$$

and uses this to compute the Tor functors. This is where the relevance of Lie $\mathbb{K}$ -algebra cohomology comes from, but we find the categorical introduction better motivates the generalization to dg Lie $\mathbb{K}$ -algebras. However it poorly motivates the more tame case of Lie $\mathbb{K}$ -algebras, so we include this remark.

Definition 3.1.6. A $\mathfrak{g}$ -module is a $\mathbb{K}$ -vector space M together with a multiplication map $\mathfrak{g} \otimes_{\mathbb{K}} M \rightarrow M$. [redo this with operads later then continue on to define Lie $\mathbb{K}$ -algebra (co)homology with values in a $\mathfrak{g}$ -module]

3.2 Differential graded Lie $\mathbb{K}$ -algebra (co)homology

Let $(\mathfrak{g}, d_{\mathfrak{g}}) = (\mathfrak{g}^{\bullet}, d_{\mathfrak{g}}^{\bullet})$ be a dg Lie $\mathbb{K}$ -algebra. We would now like to construct an analog of definition 3.1.1 for $\mathfrak{g}$.

Throughout this section, degree considerations will play a significant role, so we adopt the following conventions. Let $s : \text{Ch}(\mathbb{K}) \rightarrow \text{Ch}(\mathbb{K})$ denote the *suspension functor* $sC = C[1]$. Note that $C^k = sC^{k-1}$, so for any homogeneous element $x \in C^k$, denote by sx the corresponding homogeneous element in sC^{k-1} . Note that $|sx| = |x| - 1$. We denote by $s^{-1} : \text{Ch}(\mathbb{K}) \rightarrow \text{Ch}(\mathbb{K})$ the inverse $s^{-1}C = C[-1]$, and similarly denote by $s^{-1}x$ the homogeneous element of $s^{-1}C^{k+1}$ corresponding to $x \in C^k$.

Definition 3.2.1. The *Chevalley-Eilenberg chain complex* of $\mathfrak{g}$ is defined as $C_{\bullet}(\mathfrak{g}) := \text{Sym}^{\bullet}(\mathfrak{g}[1])$. The *Chevalley-Eilenberg cochain complex* of $\mathfrak{g}$ is defined as $C^{\bullet}(\mathfrak{g}) = \underline{\text{Hom}}_{\mathbb{K}}(C_{\bullet}(\mathfrak{g}), \mathbb{K})$.

Discussion 3.2.2. We start by investigating the Chevalley-Eilenberg chains. We do not obtain a nice description of the symmetric algebra other than the fact that multiplication satisfies Koszul braiding. Let $x, y \in \text{Sym}(\mathfrak{g}[1])$ be homogeneous elements. Then $xy = (-1)^{|x||y|}yx$. It is important to note here that if $x, y \in \mathfrak{g}$, then their homological degree is 1 (or -1? idk), not zero.

The differential on $C_{\bullet}(\mathfrak{g})$ is defined as $d_{\mathfrak{g}[1]} + d_{\text{CE}}$. Due to the degree shift, we spell out explicitly how to arrive at the formula for these. By discussion 2.1.6, we have $d_{\mathfrak{g}[1]} = -d_{\mathfrak{g}}[1]$. Let $x \in \mathfrak{g}^k$ be a homogeneous element of degree k . Write $\partial_0 = d_{\mathfrak{g}[1]}$, such that we have $\partial_0^{k-1}(sx) = -sd_{\mathfrak{g}}^k(x)$. We can extend this to a map on $C_{\bullet}(\mathfrak{g})$ by simply requiring that it be a derivation: if $x_1, \dots, x_p \in \mathfrak{g}$ are homogeneous elements whose degrees in $C_{\bullet}(\mathfrak{g})$ sum to k , then

$$\begin{aligned} \partial_0^k(sx_1 \cdots sx_p) &= \sum_{i=1}^p (-1)^{\sum_{j=1}^{i-1} |sx_j|} sx_1 \cdots \partial_0^{|sx_i|}(sx_i) \cdots sx_p \\ &= - \sum_{i=1}^p (-1)^{\sum_{j=1}^{i-1} |sx_j|} sx_1 \cdots sd_{\mathfrak{g}}^{|x_i|}(x_i) \cdots sx_p. \end{aligned}$$

Now we explore the Chevalley-Eilenberg differential on $C_\bullet(\mathfrak{g})$. On pairs of elements, this should be defined as the dg Lie bracket, but we must take care to track degrees. In particular, define

$$\partial_1 := s \circ [-, -] \circ (s^{-1} \otimes s^{-1}).$$

Care must be taken with the degrees such that this is well-defined. Given $sx, sy \in C^\bullet(\mathfrak{g})$ homogeneous elements, it might appear we want to define $\partial_1(sx \cdot sy) = s[x, y]$. However, recall from discussion 1.1.14 that since $s^{-1} : \mathfrak{g}[1] \rightarrow \mathfrak{g}$ has cohomological degree -1 , we define

$$(s^{-1} \otimes s^{-1})(sx \otimes sy) = (-1)^{|s^{-1}||sx|} s^{-1}(sx) s^{-1}(sy) = (-1)^{-(|x|-1)} x \otimes y = (-1)^{|x|-1} x \otimes y.$$

Thus we actually define

$$\partial_1(sx \cdot sy) = (-1)^{|x|-1} s[x, y].$$

The natural global extension is by defining

$$\partial_1(sx_1 \cdots sx_k) = \sum_{1 \leq i < j \leq k} (-1)^{|x_i||x_j|} \partial_1(sx_i \cdot sx_j) \cdot sx_1 \cdots \widehat{sx_i} \cdots \widehat{sx_j} \cdots sx_k.$$

The final differential $d = \partial_0 + \partial_1$ is then as follows. let $x_1, \dots, x_k \in \mathfrak{g}$ be homogeneous elements. Then

$$\begin{aligned} d(sx_1 \cdots sx_k) &= - \sum_{i=1}^k (-1)^{\sum_{j=1}^{i-1} |sx_j|} sx_1 \cdots sd_{\mathfrak{g}}(x_i) \cdots sx_k \\ &\quad + \sum_{1 \leq i < j \leq k} (-1)^{|x_i||x_j|} (-1)^{|x_i|-1} s[x_i, x_j] \cdots sx_1 \cdots \widehat{sx_i} \cdots \widehat{sx_j} \cdots sx_k. \end{aligned}$$

Once again, the Chevalley-Eilenberg chains do not form a dg $\mathbb{K}$ -algebra, and only form a dg $\mathbb{K}$ -coalgebra. This is weird to work with, so we prefer to work with the Chevalley-Eilenberg cochains.

Discussion 3.2.3. We investigate the underlying graded $\mathbb{K}$ -vector space of the Chevalley-Eilenberg cochains. Recall by ?? that we define $\mathfrak{g}^* := \underline{\text{Hom}}_{\mathbb{K}}(\mathfrak{g}, \mathbb{K})$. Since $\mathbb{K}$ is concentrated in degree 0, we can appeal to the explicit construction in remark 2.1.7 to recover

$$(\mathfrak{g}^*)^i = \text{Hom}_{\mathbb{K}}(\mathfrak{g}, \mathbb{K}[i]).$$

Since $\mathbb{K}[i]$ is concentrated in degree $-i$, a graded $\mathbb{K}$ -vector space morphism $\mathfrak{g} \rightarrow \mathbb{K}[i]$ is precisely a $\mathbb{K}$ -linear map $\mathfrak{g}^{-i} \rightarrow \mathbb{K}$. Thus we obtain

$$(\mathfrak{g}^*)^i \cong \text{Hom}_{\mathbb{K}}(\mathfrak{g}^{-i}, \mathbb{K}) \cong (\mathfrak{g}^{-i})^*.$$

Then of course $(\mathfrak{g}^*[-1])^i \cong (\mathfrak{g}^{1-i})^*$, meaning $C^\bullet(\mathfrak{g}) \cong \text{Sym}(\mathfrak{g}^*[-1])$. Following ?? then tells us what homogeneous elements of the symmetric algebra are. A homogeneous element $\omega \in C^\bullet(\mathfrak{g})$ of degree k is any $\mathbb{K}$ -linear combination of products of homogeneous elements $x_1, \dots, x_p \in \mathfrak{g}^*$ such that

$$\sum_{1 \leq i \leq p} (|x_i| + 1) = |\omega|.$$

The factor of $+1$ comes from the shift: a homogeneous element of degree i in $\mathfrak{g}^*$ is homogeneous of degree $i+1$ in $\mathfrak{g}^*[-1]$. Hence we must have

$$\sum_{1 \leq i \leq p} |x_i| = |\omega| - p.$$

Although we no longer have the nice structure of an exterior algebra as in the non-graded case, $C^\bullet(\mathfrak{g})$ is still a dg commutative $\mathbb{K}$ -algebra, and thus discussion 1.3.8 tells us that multiplication obeys Koszul braiding

$$xy = (-1)^{|x||y|}yx$$

with respect to the degrees in $C^\bullet(\mathfrak{g})$.

Discussion 3.2.4. Now we are interested in the differential on $C^\bullet(\mathfrak{g})$. Recall by discussion 3.2.2 that we define the differential on $\text{Sym}(\mathfrak{g}[1])$ by

$$\begin{aligned} d(sx_1 \cdots sx_k) &= - \sum_{i=1}^k (-1)^{\sum_{j=1}^{i-1} |sx_j|} sx_1 \cdots sd_{\mathfrak{g}}(x_i) \cdots sx_k \\ &\quad + \sum_{1 \leq i < j \leq k} (-1)^{|x_i||x_j|} (-1)^{|x_i|-1} s[x_i, x_j] \cdots sx_1 \cdots \widehat{sx_i} \cdots \widehat{sx_j} \cdots sx_k. \end{aligned}$$

Write $d^\bullet$ for the cohomological differential and $d_\bullet$ for the homological differential. Recall from construction 2.1.8 that in $C^\bullet(\mathfrak{g}) = \underline{\text{Hom}}_{\mathbb{K}}(C_\bullet(\mathfrak{g}), \mathbb{K})$, the differential is given by

$$d^\bullet(f) = d_{\mathbb{K}} \circ f - (-1)^{|f|} f \circ d_\bullet.$$

Since $d_{\mathbb{K}} = 0$, we obtain that the differential in $C^\bullet(\mathfrak{g})$ is given by

$$d^\bullet = -(-1)^{|\cdot|} d_\bullet^*.$$

With notation as in discussion 3.2.2, write $d_\bullet = \partial_0 + \partial_1$. Since $\text{Vect}_{\mathbb{K}}$ is an abelian category, and since cochains $\omega \in C^k(\mathfrak{g})$ are $\mathbb{K}$ -linear maps $\omega : C_k(\mathfrak{g}) \rightarrow \mathbb{K}$, we obtain

$$d^\bullet \omega = -(-1)^{|\omega|} \omega \circ d_\bullet = -(-1)^{|\omega|} \omega \circ (\partial_0 + \partial_1) = -(-1)^{|\omega|} \omega \circ \partial_0 - (-1)^{|\omega|} \omega \circ \partial_1.$$

Hence we obtain that $d^\bullet$ can be described as a graded sum of two operators ∂_0^* and ∂_1^* , which we write as $d_0^\bullet = -(-1)^{|\cdot|} \partial_0^*$ and $d_1^\bullet = -(-1)^{|\cdot|} \partial_1^*$.

As always, we don't want to view n -cochains as maps $C_n(\mathfrak{g}) \rightarrow \mathbb{K}$, but rather as degree n homogeneous elements of $\text{Sym}(\mathfrak{g}^*[-1])$, as explored in discussion 3.2.3. So we explore the isomorphisms in detail to obtain a formula for taking $d_0^\bullet$ and $d_1^\bullet$ on homogeneous elements of $\text{Sym}(\mathfrak{g}^*[-1])$.

Recall that $\text{Sym}(\mathfrak{g}^*[-1])$ is a dg commutative $\mathbb{K}$ -algebra, so the differential acts as a derivation, so it suffices to define it on elements $s^{-1}\alpha$. We of course illustrate how it extends more generally. Let $\alpha : \mathfrak{g}^k \rightarrow \mathbb{K}$ a map of graded $\mathbb{K}$ -vector spaces, which determines the homogeneous element $\alpha \in C^{-k}(\mathfrak{g})$ by identifying it with the map $C_\bullet(\mathfrak{g}) \rightarrow \mathbb{K}[k]$ which is α in degree $-k$. Nope that's wrong it conflates $\mathfrak{g}$ with $\text{Sym}(\mathfrak{g})$

Discussion 3.2.5. We can now transparently recover the cochain differential $D = d_0 + d_1$ from construction 3.2.6 by dualizing the chain complex.

The cochain complex is defined as $C^\bullet(\mathfrak{g}) = \underline{\text{Hom}}_{\mathbb{K}}(C_\bullet(\mathfrak{g}), \mathbb{K})$. Let $s^{-1}\alpha \in \mathfrak{g}^*[-1]$ be a degree 1 generator of the cochains, dual to $sx \in \mathfrak{g}[1]$. The canonical pairing evaluates as $\langle s^{-1}\alpha, sx \rangle = \alpha(x)$.

The cochain differential D is the formal adjoint of the chain differential ∂ . For any cochain ω and chain c , the duality relation defines D such that:

$$\langle D\omega, c \rangle = -(-1)^{|\omega|} \langle \omega, \partial c \rangle.$$

Dualizing ∂_0 to recover d_0 : Let $\omega = s^{-1}\alpha \in \mathfrak{g}^*[-1]$, which has degree $|\omega| = 1 - |\alpha|$. Applying the duality relation to a chain sx :

$$\begin{aligned}\langle d_0(s^{-1}\alpha), sx \rangle &= -(-1)^{1-|\alpha|} \langle s^{-1}\alpha, \partial_0(sx) \rangle \\ &= (-1)^{-|\alpha|} \langle s^{-1}\alpha, s(d_{\mathfrak{g}}x) \rangle \\ &= -(-1)^{|\alpha|} \alpha(d_{\mathfrak{g}}x).\end{aligned}$$

This exactly recovers the definition of d_0 constructed in construction 3.2.6.

Dualizing ∂_1 to recover d_1 : Let $\omega = s^{-1}\alpha \in \mathfrak{g}^*[-1]$. We pair $d_1(s^{-1}\alpha)$ against the 2-chain $sx \cdot sy$. Using the duality relation:

$$\begin{aligned}\langle d_1(s^{-1}\alpha), sx \cdot sy \rangle &= -(-1)^{1-|\alpha|} \langle s^{-1}\alpha, \partial_1(sx \cdot sy) \rangle \\ &= (-1)^{-|\alpha|} \langle s^{-1}\alpha, (-1)^{|x|-1} s([x, y]) \rangle \\ &= (-1)^{|x|-|\alpha|-1} \alpha([x, y]).\end{aligned}$$

If we evaluate the specific case where x and y are both degree 0 (reducing to a regular Lie algebra), the sign becomes $(-1)^{-|\alpha|-1} = -(-1)^{-|\alpha|}$. Evaluated on a degree 0 functional where $|\alpha| = 0$, this collapses precisely to $-\alpha([x, y])$, fully reconciling the minus sign derived in the non-DGLA case.

Construction 3.2.6. Let $(\mathfrak{g}, d_{\mathfrak{g}})$ be a DGLA. We construct the Chevalley-Eilenberg cochain complex $C^\bullet(\mathfrak{g})$.

The underlying graded vector space is identical to the regular Lie algebra case: we shift the dual space and take the free graded commutative algebra, yielding

$$C^\bullet(\mathfrak{g}) = \text{Sym}(\mathfrak{g}^*[-1]).$$

As established previously, the shift and the Koszul braiding canonically identify this space with $\Lambda^\bullet \mathfrak{g}^*$ as a graded vector space. However, we must now define a total differential D of cohomological degree +1 that incorporates both the internal differential $d_{\mathfrak{g}}$ and the Lie bracket.

We define D as the sum of two operators: $D = d_0 + d_1$.

The internal differential (d_0): The internal differential $d_{\mathfrak{g}} : \mathfrak{g} \rightarrow \mathfrak{g}[1]$ dualizes to a map $d_{\mathfrak{g}}^* : \mathfrak{g}^* \rightarrow \mathfrak{g}^*[1]$. Applying the shift functor $[-1]$ to both domain and codomain yields a map of degree +1 on the generators:

$$d_0 : \mathfrak{g}^*[-1] \rightarrow \mathfrak{g}^*[-1].$$

For a generator $s^{-1}\alpha \in \mathfrak{g}^*[-1]$ and $x \in \mathfrak{g}$, this evaluates as $d_0(s^{-1}\alpha)(x) = -(-1)^{|\alpha|} \alpha(d_{\mathfrak{g}}(x))$. Because $\text{Sym}(\mathfrak{g}^*[-1])$ is a free algebra, d_0 extends uniquely to a degree +1 graded derivation on the entire algebra. Crucially, d_0 preserves the symmetric power (the "word length") of the cochains.

The Chevalley-Eilenberg differential (d_1): The bracket $[-, -] : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{g}$ dualizes and shifts exactly as derived in the regular Lie algebra case, yielding a map of degree +1 on the generators:

$$d_1 : \mathfrak{g}^*[-1] \rightarrow \text{Sym}^2(\mathfrak{g}^*[-1]).$$

This also extends uniquely to a degree +1 graded derivation on the entire algebra. Unlike d_0 , d_1 increases the symmetric power by 1.

Proposition 3.2.7. *The total differential $D = d_0 + d_1$ satisfies $D^2 = 0$, making $(C^\bullet(\mathfrak{g}), D)$ a valid differential graded commutative algebra.*

Proof. Because D is the sum of two degree +1 graded derivations, its square expands as:

$$D^2 = (d_0 + d_1)^2 = d_0^2 + (d_0 \circ d_1 + d_1 \circ d_0) + d_1^2.$$

Since d_0 increases the word length by 0 and d_1 increases it by 1, the three terms on the right-hand side increase the word length by 0, 1, and 2, respectively. For the sum to be identically zero, each component must vanish independently.

1. $d_0^2 = 0$: This follows directly from the fact that $d_{\mathfrak{g}}^2 = 0$ on the underlying module $\mathfrak{g}$.
2. $d_1^2 = 0$: As verified in the case of regular Lie algebras, this condition is exactly equivalent to the graded Jacobi identity for the bracket $[-, -]$.
3. $d_0 \circ d_1 + d_1 \circ d_0 = 0$: This is the graded commutator of the two derivations. Evaluated on a dual generator, this vanishing condition translates precisely to the requirement that $d_{\mathfrak{g}}$ acts as a graded derivation over the Lie bracket (axiom 3 of definition [2.3.1](#)).

Thus, $D^2 = 0$, and the cohomology $H^\bullet(\mathfrak{g}) = \ker(D)/\text{im}(D)$ is well-defined. ■

Chapter 4

Bundles

An important class of graded objects to consider is vector and fiber bundles. A graded bundle has a sheaf of sections with values in cochain complexes. Throughout this section, $\mathbb{K}$ denotes either $\mathbb{R}$ or $\mathbb{C}$.

4.1 Graded bundles

Definition 4.1.1. Let X be a topological space. A *fiber bundle* over X is a tuple (E, F, π, X) , where $\pi : E \rightarrow X$ is a surjection, and for all $x \in X$, there is an open neighborhood $U \subseteq X$ of x and a homeomorphism $\varphi : \pi^{-1}(U) \rightarrow U \times F$ such that the diagram

$$\begin{array}{ccc} E & \xrightarrow{\varphi} & U \times F \\ \downarrow \pi & \swarrow \pi_U & \\ U & & \end{array}$$

commutes, where $\pi_U(u, f) = u$ is the natural projection of the product space. If X and E are smooth manifolds, then the fiber bundle is *smooth* when π is smooth and the maps φ are diffeomorphisms. If X and E are complex manifolds, then the fiber bundle is *holomorphic* if π is holomorphic and the maps φ are biholomorphisms. We generally refer to E as the (total space of the) fiber bundle, X as the base space, π as the projection, and F as the fiber. We call the maps φ *local trivializations*.

Definition 4.1.2. Let $\mathbb{K}$ be a topological field and X a topological space. A $\mathbb{K}$ -*vector bundle* is a fiber bundle $\pi : E \rightarrow X$ where the fiber is $\mathbb{K}^n$ for some $n \in \mathbb{N}$. If $\mathbb{K}$ is a smooth or holomorphic manifold, then a $\mathbb{K}$ -vector bundle is smooth or holomorphic when it is smooth or holomorphic as a fiber bundle. The number n is called the *rank* of the vector bundle. If the field $\mathbb{K}$ is understood, one often refers to n -vector bundles as $\mathbb{K}$ -vector bundles of rank n .

Remark 4.1.3. One could develop a theory of R -module bundles, but I see no reason to do so, since it is already difficult to develop the theory of $\mathbb{K}$ -vector bundles for arbitrary $\mathbb{K}$. For example, as noted in ??, arbitrary fields $\mathbb{K}$ need not obtain the structure of an $\mathcal{O}_X$ -module, which severely restricts their theory. Hence, we adopt the following convention.

If the base space is a *smooth* manifold, then we only consider $\mathbb{K}$ -vector bundles where $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$. If the base space is a *complex* manifold, we will only consider $\mathbb{C}$ -vector bundles. From now on, whenever we specify a $\mathbb{K}$ -vector bundle $\pi : E \rightarrow X$, we implicitly assume the vector bundle is either smooth or complex, and that the field $\mathbb{K}$ aligns with this remark. We do not work with general $\mathbb{K}$ -vector bundles over arbitrary topological spaces. We will, however, work with fiber bundles over arbitrary topological spaces.

Discussion 4.1.4. Let X be a topological space and $\pi : E \rightarrow X$ a fiber bundle with fiber F . For all $x \in X$, we denote by E_x the fiber $\pi^{-1}(x)$ of the projection. Choose an open neighborhood U of x , and a local trivialization $\varphi : \pi^{-1}(U) \rightarrow U \times F$. Commutativity of the diagram in definition 4.1.1 implies that $\varphi(\pi^{-1}(x)) = \{x\} \times F$, since we must have that $\pi(\varphi(\pi^{-1}(x))) = \pi(\pi^{-1}(x)) = x$. Hence each fiber E_x can be identified with the fiber F . This justifies the terminology “fiber bundle”, since one thinks of E as many copies of the fiber F bundled in a topological way.

Discussion 4.1.5. By the same logic as discussion 4.1.4, if $\pi : E \rightarrow X$ is a $\mathbb{K}$ -vector bundle, then each fiber E_x of $x \in X$ can be identified with $\{x\} \times \mathbb{K}^n \cong \mathbb{K}^n$. Note that this is a vector space, hence justifying the terminology “vector bundle”.

Definition 4.1.6. Let $\pi : E \rightarrow X$ be a vector bundle. A *section* of the vector bundle over an open set $U \subseteq X$ is a continuous map $s : U \rightarrow E$ such that $\pi \circ s = 1$, for 1 the identity map. The collection of sections of E over U is denoted $\Gamma(U, E)$. Sections over X are called *global sections*. For all $x \in X$, we write $s_x = s(x)$. Whenever E is smooth or holomorphic, we speak of smooth or holomorphic sections. Denote smooth sections of a smooth vector bundle E by $\Gamma^\infty(-, E)$, and holomorphic sections of a holomorphic vector bundle E by $\Gamma^0(-, E)$.

Discussion 4.1.7. Let $\pi : E \rightarrow X$ be a $\mathbb{K}$ -vector bundle of rank n . Let $s : U \rightarrow E$ be a section over some open set $U \subseteq X$. The section condition implies that for all $x \in X$, we have $\pi(s(x)) = x$. In particular, $s_x \in E_x$. Note that since $E_x \cong \mathbb{K}^n$ can be viewed as a vector space, we can view s_x as a *vector* $s_x \in \mathbb{K}^n$. Thus a section s_x is a map which assigns to each $x \in X$ a vector $s_x \in E_x$.

Example 4.1.8. Let M be an n -manifold. Then TM is a $\mathbb{R}$ -vector bundle of rank n , and sections are vector fields. Similarly, $\Lambda^k T^*M$ is a $\mathbb{R}$ -vector bundle over M , and sections are differential k -forms. These are the canonical examples of vector bundles, and motivates their existence as objects which allow us to construct various types of vector fields.

Definition 4.1.9. Let $\pi_1 : E_1 \rightarrow X$ and $\pi_2 : E_2 \rightarrow X$ be fiber bundles on a topological space X with fiber F . Then a morphism $\varphi : E_1 \rightarrow E_2$ of fiber bundles is a continuous map $\varphi : E_1 \rightarrow E_2$ which commutes with the projections in the sense that the diagram

$$\begin{array}{ccc} E_1 & \xrightarrow{\varphi} & E_2 \\ & \searrow \pi_1 & \swarrow \pi_2 \\ & X & \end{array}$$

commutes. One obtains a category $\text{Fib}_F(X)$ of fiber bundles over X with fiber F . If the fiber bundles are smooth or holomorphic, then one speaks of smooth or holomorphic morphisms, and obtains categories $\text{Fib}_F^\infty(X)$ and $\text{Fib}_F^0(X)$ of smooth and holomorphic fiber bundles, respectively. One similarly obtains categories $\text{Vect}_{\mathbb{K}}(X)$, $\text{Vect}_{\mathbb{K}}^\infty(X)$, and $\text{Vect}_{\mathbb{K}}^0(X)$ of $\mathbb{K}$ -vector bundles.

Warning 4.1.10. The categories $\text{Fib}_{\mathbb{K}}^\infty(X)$ and $\text{Fib}_F^0(X)$ need not be subcategories of $\text{Fib}_F(X)$, or each other. For example, many smooth vector bundles admit many distinct holomorphic structures that promote them to holomorphic vector bundles.

Definition 4.1.11. A *graded $\mathbb{K}$ -vector bundle* E over X is a functor $E : \mathbb{Z} \rightarrow \text{Vect}_{\mathbb{K}}(X)$. More generally, a *graded fiber bundle* is a map $E : \mathbb{Z} \rightarrow \text{Fib}_F(X)$. We also speak of smooth and holomorphic fiber bundles.

Notation 4.1.12. As before, if E is a graded vector bundle, then we denote $E(i)$ by E^i , and write the projection as π^i . Hence E can be presented as a collection $\{\pi^i : E^i \rightarrow X\}_{i \in \mathbb{Z}}$.

Discussion 4.1.13. The category $\text{Vect}_{\mathbb{K}}(X)$, and the smooth and holomorphic categories, are *not abelian categories*. However, they have many universals, and are computed fiberwise. Important examples include the direct sum

$$(E \oplus F)_x := E_x \oplus F_x,$$

tensor product

$$(E \otimes_{\mathbb{K}} F)_x := E_x \otimes_{\mathbb{K}} F_x,$$

and the dual

$$(E^*)_x := (E_x)^*.$$

We can also speak of the kernel and cokernel of a map $\varphi : E \rightarrow F$ of $\mathbb{K}$ -vector bundles by defining

$$\text{Ker } \varphi := \{x \in E \mid \varphi(x) = 0\},$$

where 0 lives in the appropriate fiber of F . Similarly, the cokernel is the quotient of F by the set-theoretic image of φ . The reason these categories are not abelian is that these (co)kernels need not be vector bundles in general, and the issue arises from the fact that E and F could have different ranks, meaning the fibers of the resultant object could have a non-constant rank. However, one can still speak of the kernel and cokernel as topological spaces.

Construction 4.1.14. To fix the issue of vector bundles not forming abelian categories, we pass to the theory of sheaves. Let $\pi : E \rightarrow X$ be a $\mathbb{K}$ -vector bundle. If $U \subseteq X$ is open, then the sections $\Gamma(U, E)$ form a vector space: if s, t are sections of E over U , then by definition, for all $x \in U$ we have $s_x, t_x \in E_x$. Since E_x is a vector space, $(s + t)_x = s_x + t_x \in E_x$. Similarly, for all $c \in \mathbb{R}$, we have $(cs)_x = c(s_x) \in E_x$. The other axioms are similarly checked pointwise. Therefore $\Gamma(U, E)$ is a vector space.

Additionally, let $U \subseteq V$ be open in X . Let $s \in \Gamma(V, E)$. Then the restriction $s|_U$ is manifestly a section $s|_U \in \Gamma(U, E)$, yielding a restriction map $|_U : \Gamma(V, E) \rightarrow \Gamma(U, E)$. It is trivial to show that restriction is $\mathbb{K}$ -linear. Thus we obtain a presheaf $\Gamma(-, E)$ valued in $\mathbb{K}$ -vector spaces. If E is a vector bundle, we generally denote the presheaf $\Gamma(U, E)$ by $\mathcal{E}(U)$, using a cursive font.

We claim that $\mathcal{E}$ is a sheaf. Indeed, let $\{U_i\}_{i \in I}$ be an open cover of some open set $U \subseteq X$, and let $s_i : U_i \rightarrow E$ be sections. It suffices to show that there is a unique section $s : U \rightarrow E$ with $s|_{U_i} = s_i$ for all $i \in I$. This is immediate for continuous, smooth, and holomorphic vector bundles by the usual gluing theorems for functions. Thus $\mathcal{E}$ is a sheaf of $\mathbb{K}$ -vector spaces, called the *sheaf of sections* of E . We obtain the same results for the sheaves of smooth sections and of holomorphic sections. Generally, we will choose to only work with one type of sections (continuous smooth, holomorphic), and will write $\mathcal{E}$ for the sheaf of those sections. Rarely will we need to consider multiple different types of sheaves of sections at once.

Remark 4.1.15. Let E be a graded $\mathbb{K}$ -vector bundle over a space X , and $x \in X$. Then the collection $E_x = \{E_x^i\}$ of fibers forms a graded $\mathbb{K}$ -vector space. This suggests a definition for a section of E .

Definition 4.1.16. Let E be a graded $\mathbb{K}$ -vector bundle over X , and $U \subseteq X$ an open set. A *section* s of E over U is a continuous map

$$s : X \rightarrow \bigoplus_{i \in \mathbb{Z}} E^i.$$

such that $s(x) = s_x \in \bigoplus_i E_x^i$. Equivalently, a section s of E is a collection of continuous maps $\{s^i : X \rightarrow E^i\}_{i \in \mathbb{Z}}$ such that $s_x^i \in E_x^i$ for all $x \in U$, all but finitely many s_x^i are zero. Thus $s = \bigoplus_i s^i$. One also speaks of smooth and holomorphic sections.

Remark 4.1.17. The definition definition 4.1.16 follows the modern intuition that a graded object should not just be a collection of objects, but rather a canonical decomposition of an object. For example, the tensor algebra $T(M)$ of a module M is a graded object, and admits a canonical decomposition into the tensor powers of M . We find it more intuitive to think of graded objects as collections of objects, but in this case it leads to a more poorly behaved definition, so we use the more common intuition.

Discussion 4.1.18. We consider the sheaf of sections $\mathcal{E}$ of a graded $\mathbb{K}$ -vector bundle $E \rightarrow X$. We obviously claim this object is a sheaf. In particular, we claim $\mathcal{E}$ is a graded $\mathcal{O}_X$ -module in the sense that $\mathcal{E}(U)$ is a graded $\mathcal{O}_X(U)$ -module, where $\mathcal{O}_X$ is the structure sheaf of X :

- If $E \rightarrow X$ is a graded holomorphic $\mathbb{K}$ -vector bundle, then we take the structure sheaf $\mathcal{O}_X(U)$ to be holomorphic functions $U \rightarrow \mathbb{C}$.
- If $E \rightarrow X$ is a graded smooth $\mathbb{K}$ -vector bundle, then we take the structure sheaf $\mathcal{O}_X(U)$ to be smooth functions $C^\infty(U)$.
- If $E \rightarrow X$ is a graded $\mathbb{K}$ -vector bundle, then we take the structure sheaf $\mathcal{O}_X(U)$ to be continuous functions $C^0(U, \mathbb{R})$.

First we show $\mathcal{E}$ is a presheaf. We write E for the direct sum $\bigoplus_i E^i$ and $\pi : E \rightarrow X$ for the induced projection, such that sections of E are sections of the projection. Let $s : V \rightarrow E$ be a section of E over some open $V \subseteq X$. If $U \subseteq V$ is another open set, then simply restricting $s|_V$ yields a section over U ; note also that $(s|_U)^i = s^i|_U$ since restriction is precomposition by the inclusion, which distributes over the direct sum.

Now we show $\mathcal{E}(U)$ is an $\mathcal{O}_X(U)$ -module. All cases proceed analogously: if we have a section of the structure sheaf $f \in \mathcal{O}_X(U)$ and a section $s \in \mathcal{E}(U)$ of E , then one defines $(f \cdot s)_x := f(x)s_x$ for all $x \in X$. Since $f(x) \in \mathbb{R}$ and $\mathcal{E}^i$ is a sheaf of $\mathbb{R}$ -vector spaces, we are done. Remark: the $\mathbb{K}$ -generality is a mess

4.2 wtf

Let $E = \{\pi^i : E^i \rightarrow M\}_{i \in \mathbb{Z}}$ be a graded vector bundle over a smooth manifold M . We like to view this data as a single vector bundle E which admits a canonical decomposition

$$E = \bigoplus_{i \in \mathbb{Z}} E^i.$$

Universality of coproduct yields a projection map $\pi : E \rightarrow M$ given by $\pi|_{E^i} = \pi^i$. This yields, for all $x \in M$,

$$\pi^{-1}(x) = \bigoplus_{i \in \mathbb{Z}} (\pi^i)^{-1}(x) = \bigoplus_{i \in \mathbb{Z}} E_x^i.$$

Hence

$$E_x = \bigoplus_{i \in \mathbb{Z}} E_x^i.$$

From this perspective, a section s of a graded vector bundle E is a section of the conventional vector bundle E , meaning a map $s : M \rightarrow E$ with the property that for all $x \in M$, writing $s(x) =: s_x$,

$$s_x \in E_x = \bigoplus_{i \in \mathbb{Z}} E_x^i.$$

Hence s_x is a collection $\{s_x^i\}_{i \in \mathbb{Z}}$ of vectors $s_x^i \in E_x^i$ such that all but finitely many s_x^i are zero.

If $E = \bigoplus_{i \in \mathbb{Z}} E_i$ is a graded vector bundle over a manifold M , I intuitively expect a section $s \in \Gamma(U, E)$ of E to be a tuple $s = \{s_i \in \Gamma(U, E_i)\}_{i \in \mathbb{Z}}$, but writing E as a direct sum implies that $(s_i)_x = 0$ for all but finitely many i , which I do not intuitively expect.

Bibliography

- [Lur17] Jacob Lurie. Higher algebra. Unpublished. Available at <https://www.math.ias.edu/~lurie/papers/HA.pdf>, September 2017.
- [ML98] Saunders Mac Lane. *Categories for the Working Mathematician*, volume 5 of *Graduate Texts in Mathematics*. Springer, second edition, 1998.